JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Vector Algebra

JEE Main Mathematics · Previous Year Questions with Answer Key

281	182	2002 – 2026	58	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	176 questions
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Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 176 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\vec{b} \times \vec{c}) = \vec{0}$.

If $\vec{a} \cdot \vec{c} = 15$, then $\vec{c} \cdot (\hat{i} + \hat{j} - 3\hat{k})$ is equal to :

(A) -6 (B) -5 (C) -4 (D) -3

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$ and $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is :

(A) 450 (B) 900 (C) 1800 (D) 2400

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Let O be the origin, $\overrightarrow{OP} = \vec{a}$ and $\overrightarrow{OQ} = \vec{b}$. If R is the point on $\overrightarrow{OP}$ such that $\overrightarrow{OP} = 5\overrightarrow{OR}$, and M is the point such that $\overrightarrow{OQ} = 5\overrightarrow{RM}$, then $\overrightarrow{PM}$ is equal to :

(A) $\frac{1}{5}(\vec{a} - 4\vec{b})$ (B) $\frac{1}{5}(\vec{b} - 4\vec{a})$ (C) $\frac{1}{5}(-\vec{a} + 4\vec{b})$ (D) $\frac{1}{5}(-\vec{b} + 4\vec{a})$

Q4. JEE Main 2026 (Online) 5th April Morning Shift

Let $\vec{a} = \sqrt{7}\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$, then $|3\vec{r}|^2$ is equal to:

(A) 44 (B) 54 (C) 86 (D) 132

Q5. JEE Main 2026 (Online) 4th April Evening Shift

Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $|\hat{u} \times \hat{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \lambda\hat{u} + \hat{v} + (\hat{u} \times \hat{v})$, then λ is equal to:

(A) $\frac{4}{3}(\vec{A} \cdot \hat{u}) - \frac{2}{3}(\vec{A} \cdot \hat{v})$

(B) $\frac{2}{3}(\vec{A} \cdot \hat{u}) - \frac{1}{3}(\vec{A} \cdot \hat{v})$

(C) $\frac{4}{3}(\vec{A} \cdot \hat{u}) + \frac{2}{3}(\vec{A} \cdot \hat{v})$

(D) $(\vec{A} \cdot \hat{u}) - \frac{1}{2}(\vec{A} \cdot \hat{v})$

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Let the vectors $\vec{a} = -\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$. For some $\lambda, \mu \in \mathbb{R}$, let $\vec{c} = \lambda\vec{a} + \mu\vec{b}$.

If $\vec{c} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 10$ and $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = -2$, then $|\vec{c}|^2$ is equal to :

(A) 8 (B) 12 (C) 14 (D) 15

Q7. JEE Main 2026 (Online) 2nd April Evening Shift

Two adjacent sides of a parallelogram PQRS are given by $\overrightarrow{PQ} = \hat{j} + \hat{k}$ and $\overrightarrow{PS} = \hat{i} - \hat{j}$. If the side PS is rotated about the point P by an acute angle α in the plane of the parallelogram so that it becomes perpendicular to the side PQ, then $\sin^2\left(\frac{5\alpha}{2}\right) - \sin^2\left(\frac{\alpha}{2}\right)$ is equal to:

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{2\sqrt{3}}{5}$

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, then the maximum value of $3|(3\vec{a} + 2\vec{b})| + 4|(3\vec{a} - 2\vec{b})|$ is :

(A) 30 (B) 36 (C) 60 (D) 72

Q9. JEE Main 2026 (Online) 28th January Evening Shift

Let P be a point in the plane of the vectors $\overrightarrow{AB} = 3\hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{AC} = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equidistant from the lines AB and AC. If $|\overrightarrow{AP}| = \frac{\sqrt{5}}{2}$, then the area of the triangle ABP is :

(A) $\frac{3}{2}$ (B) $\frac{\sqrt{30}}{4}$ (C) $\frac{\sqrt{26}}{4}$ (D) 2

Q10. JEE Main 2026 (Online) 28th January Morning Shift

For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying

$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ and $|2\vec{a} + k\vec{b} + k\vec{c}| = 3$,

the positive value of k is :

(A) 4 (B) 5 (C) 6 (D) 3

Q11. JEE Main 2026 (Online) 24th January Evening Shift

Let $\vec{a} = 2\hat{\mathrm{i}} - \hat{\mathrm{j}} - \hat{\mathrm{k}}, \vec{b} = \hat{\mathrm{i}} + 3\hat{\mathrm{j}} - \hat{\mathrm{k}}$ and $\vec{c} = 2\hat{\mathrm{i}} + \hat{\mathrm{j}} + 3\hat{\mathrm{k}}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

(A) $\frac{\sqrt{35}}{2}$ (B) 13 (C) $\frac{\sqrt{21}}{2}$ (D) 7

Q12. JEE Main 2026 (Online) 24th January Evening Shift

Let $\vec{a} = 2\hat{\mathrm{i}} - 5\hat{\mathrm{j}} + 5\hat{\mathrm{k}}$ and $\vec{b} = \hat{\mathrm{i}} - \hat{\mathrm{j}} + 3\hat{\mathrm{k}}$. If $\vec{c}$ is a vector such that $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}) \cdot \vec{c} = -97$, then $|\vec{c} \times \hat{\mathrm{k}}|^2$ is equal to

(A) 193 (B) 205 (C) 218 (D) 233

Q13. JEE Main 2026 (Online) 24th January Morning Shift

Let $\vec{a} = 2\hat{\mathrm{i}} + \hat{\mathrm{j}} - 2\hat{\mathrm{k}}, \vec{b} = \hat{\mathrm{i}} + \hat{\mathrm{j}}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let $\vec{d}$ be a vector such that $|\vec{d} - \vec{a}| = \sqrt{11}, |\vec{c} \times \vec{d}| = 3$ and the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a} \cdot \vec{d}$ is equal to

(A) 0 (B) 1 (C) 3 (D) 11

Q14. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c})$. If $|\vec{a}| = 1, |\vec{b}| = 4, |\vec{c}| = 2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a} \cdot \vec{c}|$ is equal to

(A) 2 (B) 0 (C) 1 (D) 4

Q15. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\vec{a} = \hat{\mathrm{i}} - 2\hat{\mathrm{j}} + 3\hat{\mathrm{k}}, \vec{b} = 2\hat{\mathrm{i}} + \hat{\mathrm{j}} - \hat{\mathrm{k}}, \vec{c} = \lambda\hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}}$ and $\vec{v} = \vec{a} \times \vec{b}$. If $\vec{v} \cdot \vec{c} = 11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to

(A) 9 (B) 6 (C) 4 (D) 12

Q16. JEE Main 2026 (Online) 23rd January Morning Shift

Let $\overrightarrow{\mathrm{a}} = -\hat{i} + \hat{j} + 2\hat{k}, \overrightarrow{\mathrm{b}} = \hat{i} - \hat{j} - 3\hat{k}, \overrightarrow{\mathrm{c}} = \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}$ and $\overrightarrow{\mathrm{d}} = \overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{a}}$. Then $(\vec{a} - \vec{b}) \cdot \vec{d}$ is equal to :

(A) 2 (B) -2 (C) 4 (D) -4

Q17. JEE Main 2026 (Online) 22nd January Evening Shift

Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \lambda\hat{j} + 2\hat{k}, \lambda \in \boldsymbol{Z}$ be two vectors. Let $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}| = \sqrt{53}$, then the maximum possible value of $(\vec{c} \cdot \vec{d})^2$ is equal to :

(A) 52 (B) 26 (C) 104 (D) 208

Q18. JEE Main 2026 (Online) 22nd January Morning Shift

Let $\overrightarrow{\mathrm{AB}} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\overrightarrow{\mathrm{AD}} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \lambda \in \mathbb{R}$. Let the projection of the vector $\vec{v} = \hat{i} + \hat{j} + \hat{k}$ on the diagonal $\overrightarrow{\mathrm{AC}}$ of the parallelogram ABCD be of length one unit. If α, β, where $\alpha > \beta$, be the roots of the equation $\lambda^2 x^2 - 6\lambda x + 5 = 0$, then $2\alpha - \beta$ is equal to

(A) 3 (B) 6 (C) 4 (D) 1

Q19. JEE Main 2026 (Online) 21st January Evening Shift

For a triangle ABC, let $\vec{p} = \overrightarrow{BC}, \vec{q} = \overrightarrow{CA}$ and $\vec{r} = \overrightarrow{BA}$. If $|\vec{p}| = 2\sqrt{3}, |\vec{q}| = 2$ and $\cos\theta = \frac{1}{\sqrt{3}}$, where θ is the angle between $\vec{p}$ and $\vec{q}$, then $|\vec{p} \times (\vec{q} - 3\vec{r})|^2 + 3|\vec{r}|^2$ is equal to :

(A) 200 (B) 220 (C) 410 (D) 340

Q20. JEE Main 2026 (Online) 21st January Morning Shift

Let $\overrightarrow{\mathrm{a}} = -\hat{i} + 2\hat{j} + 2\hat{k}, \overrightarrow{\mathrm{b}} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\overrightarrow{\mathrm{c}}$ be a vector such that $\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{c}} = \overrightarrow{\mathrm{b}}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

(A) 30 (B) 33 (C) 27 (D) 35

Q21. JEE Main 2026 (Online) 21st January Morning Shift

Let $\overrightarrow{c}$ and $\overrightarrow{d}$ be vectors such that $|\overrightarrow{c}+\overrightarrow{d}|=\sqrt{29}$ and $\overrightarrow{c}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\overrightarrow{d}$. If $\lambda_1,\lambda_2\,(\lambda_1>\lambda_2)$ are the possible values of $(\vec{c}+\vec{d})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$, then the equation $\mathrm{K}^2x^2+(\mathrm{K}^2-5\,\mathrm{K}+\lambda_1)xy+\left(3\,\mathrm{K}+\frac{\lambda_2}{2}\right)y^2-8x+12y+\lambda_2=0$ represents a circle, for K equal to :

(A) 4 (B) -1 (C) 2 (D) 1

Q22. JEE Main 2025 (Online) 8th April Evening Shift

Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$. Let $\hat{c}$ be a unit vector in the plane of the vectors $\vec{a}$ and $\vec{b}$ and be perpendicular to $\vec{a}$. Then such a vector $\hat{c}$ is:

(A) $\frac{1}{\sqrt{2}}(-\hat{i}+\hat{k})$ (B) $\frac{1}{\sqrt{5}}(\hat{j}-2\hat{k})$

(C) $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$ (D) $\frac{1}{\sqrt{3}}(-\hat{i}+\hat{j}-\hat{k})$

Q23. JEE Main 2025 (Online) 7th April Evening Shift

Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\frac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|}=\sqrt{2}+1$. Then $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ is :

(A) $2+\sqrt{2}$ (B) $2+4\sqrt{2}$ (C) $4+2\sqrt{2}$ (D) $1+\sqrt{2}$

Q24. JEE Main 2025 (Online) 7th April Morning Shift

Let the angle $\theta, 0<\theta<\frac{\pi}{2}$ between two unit vectors $\hat{a}$ and $\hat{b}$ be $\sin^{-1}\left(\frac{\sqrt{65}}{9}\right)$. If the vector $\vec{c}=3\hat{a}+6\hat{b}+9(\hat{a}\times\hat{b})$, then the value of $9(\vec{c}\cdot\hat{a})-3(\vec{c}\cdot\hat{b})$ is

(A) 31 (B) 29 (C) 24 (D) 27

Q25. JEE Main 2025 (Online) 4th April Morning Shift

Consider two vectors $\vec{u}=3\hat{i}-\hat{j}$ and $\vec{v}=2\hat{i}+\hat{j}-\lambda\hat{k},\lambda>0$. The angle between them is given by $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$. Let $\vec{v}=\vec{v}_1+\overrightarrow{v_2}$, where $\vec{v}_1$ is parallel to $\vec{u}$ and $\overrightarrow{v_2}$ is perpendicular to $\vec{u}$. Then the value of $\left|\overrightarrow{v_1}\right|^2+\left|\overrightarrow{v_2}\right|^2$ is equal to

(A) $\frac{23}{2}$ (B) $\frac{25}{2}$ (C) 10 (D) 14

Q26. JEE Main 2025 (Online) 2nd April Evening Shift

Let $\overrightarrow{a}=2\hat{i}-3\hat{j}+\hat{k}$, $\overrightarrow{b}=3\hat{i}+2\hat{j}+5\hat{k}$ and a vector $\overrightarrow{c}$ be such that $(\vec{a}-\vec{c})\times\vec{b}=-18\hat{i}-3\hat{j}+12\hat{k}$ and $\vec{a}\cdot\vec{c}=3$. If $\vec{b}\times\vec{c}=\vec{d}$, then $|\vec{a}\cdot\vec{d}|$ is equal to :

(A) 15 (B) 18 (C) 12 (D) 9

Q27. JEE Main 2025 (Online) 2nd April Morning Shift

If $\overrightarrow{a}$ is a nonzero vector such that its projections on the vectors $2\hat{i}-\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\overrightarrow{a}$ is :

(A) $\frac{1}{\sqrt{155}}(-7\hat{i}+9\hat{j}+5\hat{k})$ (B) $\frac{1}{\sqrt{155}}(-7\hat{i}+9\hat{j}-5\hat{k})$

(C) $\frac{1}{\sqrt{155}}(7\hat{i}+9\hat{j}-5\hat{k})$ (D) $\frac{1}{\sqrt{155}}(7\hat{i}+9\hat{j}+5\hat{k})$

Q28. JEE Main 2025 (Online) 29th January Evening Shift

Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$ and $\vec{c}=2\hat{i}+3\hat{j}-\hat{k}$, and $\hat{a}$ makes an angle of $\cos^{-1}\left(-\frac{1}{3}\right)$ with the vector $\hat{i}+\hat{j}+\hat{k}$. If $\hat{a}$ makes an angle of $\frac{\pi}{3}$ with the vector $\hat{i}+\alpha\hat{j}+\hat{k}$, then the value of α is:

(A) $\sqrt{3}$ (B) $\sqrt{6}$ (C) $-\sqrt{6}$ (D) $-\sqrt{3}$

Q29. JEE Main 2025 (Online) 29th January Morning Shift

Let $\vec{a}=2\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}-5\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}\times\vec{c}=\vec{c}\times\vec{b}$ and $(\vec{a}+\vec{c})\cdot(\vec{b}+\vec{c})=168$. Then the maximum value of $|\vec{c}|^2$ is :

(A) 77 (B) 154 (C) 308 (D) 462

Q30. JEE Main 2025 (Online) 28th January Evening Shift

If the components of $\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ along and perpendicular to $\vec{b}=3\hat{i}+\hat{j}-\hat{k}$ respectively, are $\frac{16}{11}(3\hat{i}+\hat{j}-\hat{k})$ and $\frac{1}{11}(-4\hat{i}-5\hat{j}-17\hat{k})$, then $\alpha^2+\beta^2+\gamma^2$ is equal to :

(A) 16 (B) 23 (C) 26 (D) 18

Q31. JEE Main 2025 (Online) 28th January Evening Shift

Let A, B, C be three points in xy-plane, whose position vectors are given by $\sqrt{3}\hat{i}+\hat{j}, \hat{i}+\sqrt{3}\hat{j}$ and $a\hat{i}+(1-a)\hat{j}$ respectively with respect to the origin O . If the distance of the point C from the line bisecting the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\frac{9}{\sqrt{2}}$, then the sum of all the possible values of a is :

(A) 2 (B) 0 (C) $\frac{9}{2}$ (D) 1

Q32. JEE Main 2025 (Online) 24th January Evening Shift

Let the position vectors of three vertices of a triangle be $4\vec{p}+\vec{q}-3\vec{r}$, $-5\vec{p}+\vec{q}+2\vec{r}$ and $2\vec{p}-\vec{q}+2\vec{r}$. If the position vectors of the orthocenter and the circumcenter of the triangle are $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ and $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ respectively, then $\alpha+2\beta+5\gamma$ is equal to :

(A) 4 (B) 3 (C) 1 (D) 6

Q33. JEE Main 2025 (Online) 24th January Evening Shift

Let $\overrightarrow{a}=3\hat{i}-\hat{j}+2\hat{k}$, $\overrightarrow{b}=\overrightarrow{a}\times(\hat{i}-2\hat{k})$ and $\overrightarrow{c}=\overrightarrow{b}\times\hat{k}$. Then the projection of $\overrightarrow{c}-2\hat{j}$ on $\vec{a}$ is :

(A) $2\sqrt{7}$ (B) $3\sqrt{7}$ (C) $\sqrt{14}$ (D) $2\sqrt{14}$

Q34. JEE Main 2025 (Online) 24th January Morning Shift

Let $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}, \vec{b}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$. If the vector $\vec{c}$ is perpendicular to $\vec{b}$ and $\vec{a}\cdot\vec{c}=5$, then $|\vec{c}|$ is equal to

(A) $\sqrt{\frac{11}{6}}$ (B) $\frac{1}{3\sqrt{2}}$ (C) 18 (D) 16

Q35. JEE Main 2025 (Online) 23rd January Evening Shift

Let the point A divide the line segment joining the points $\mathrm{P}(-1,-1,2)$ and $\mathrm{Q}(5,5,10)$ internally in the ratio $r:1(r>0)$. If O is the origin and $(\overrightarrow{\mathrm{OQ}}\cdot\overrightarrow{\mathrm{OA}})-\frac{1}{5}|\overrightarrow{\mathrm{OP}}\times\overrightarrow{\mathrm{OA}}|^2=10$, then the value of r is :

(A) $\sqrt{7}$ (B) 14 (C) 7 (D) 3

Q36. JEE Main 2025 (Online) 23rd January Morning Shift

Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $\hat{\mathbf{i}}+2\hat{\mathbf{j}}+\hat{\mathrm{k}}, \hat{\mathrm{i}}+3\hat{\mathrm{j}}-2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median line segment through A of the triangle ABC at the point E. If the length of AD is $\frac{\sqrt{110}}{3}$ and the volume of the tetrahedron is $\frac{\sqrt{805}}{6\sqrt{2}}$, then the position vector of E is

(A) $\frac{1}{6}(7\hat{\mathrm{i}}+12\hat{\mathrm{j}}+\hat{\mathrm{k}})$ (B) $\frac{1}{12}(7\hat{\mathrm{i}}+4\hat{\mathrm{j}}+3\hat{\mathrm{k}})$

(C) $\frac{1}{6}(12\hat{i}+12\hat{j}+\hat{k})$ (D) $\frac{1}{2}(\hat{i}+4\hat{j}+7\hat{k})$

Q37. JEE Main 2025 (Online) 23rd January Morning Shift

Let the arc AC of a circle subtend a right angle at the centre O. If the point B on the arc AC, divides the arc AC such that $\frac{\text{length of arc } AB}{\text{length of arc } BC}=\frac{1}{5}$, and $\overrightarrow{OC}=\alpha\overrightarrow{OA}+\beta\overrightarrow{OB}$, then $\alpha+\sqrt{2}(\sqrt{3}-1)\beta$ is equal to

(A) $2\sqrt{3}$ (B) $5\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $2-\sqrt{3}$

Q38. JEE Main 2025 (Online) 22nd January Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{3}$. If $\lambda\vec{a}+2\vec{b}$ and $3\vec{a}-\lambda\vec{b}$ are perpendicular to each other, then the number of values of λ in $[-1,3]$ is :

(A) 1 (B) 3 (C) 2 (D) 0

Q39. JEE Main 2024 (Online) 9th April Evening Shift

Between the following two statements:

Statement I : Let $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a}\times\vec{r}=\vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{r}=0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC, $\cos 2A+\cos 2B+\cos 2C\geq -\frac{3}{2}$.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q40. JEE Main 2024 (Online) 9th April Evening Shift

Let $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k},\vec{b}=-\hat{i}+\hat{k},\vec{c}=\beta\hat{j}-\hat{k}$, where α and β are integers and $\alpha\beta=-6$. Let the values of the ordered pair (α,β), for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1,β_1) and (α_2,β_2). Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to

(A) 21 (B) 24 (C) 19 (D) 17

Q41. JEE Main 2024 (Online) 9th April Morning Shift

Let three vectors $\overrightarrow{a}=\alpha\hat{i}+4\hat{j}+2\hat{k},\overrightarrow{b}=5\hat{i}+3\hat{j}+4\hat{k},\overrightarrow{c}=x\hat{i}+y\hat{j}+z\hat{k}$ form a triangle such that $\vec{c}=\vec{a}-\vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to:

(A) 14 (B) 12 (C) 16 (D) 10

Q42. JEE Main 2024 (Online) 9th April Morning Shift

Let $\overrightarrow{OA}=2\vec{a},\overrightarrow{OB}=6\vec{a}+5\vec{b}$ and $\overrightarrow{OC}=3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral $OABC$ is equal to:

(A) 32 (B) 38 (C) 35 (D) 40

Q43. JEE Main 2024 (Online) 8th April Evening Shift

Let $\overrightarrow{a}=4\hat{i}-\hat{j}+\hat{k},\overrightarrow{b}=11\hat{i}-\hat{j}+\hat{k}$ and $\overrightarrow{c}$ be a vector such that $(\overrightarrow{a}+\overrightarrow{b})\times\overrightarrow{c}=\overrightarrow{c}\times(-2\overrightarrow{a}+3\overrightarrow{b})$. If $(2\vec{a}+3\vec{b})\cdot\vec{c}=1670$, then $|\vec{c}|^2$ is equal to:

(A) 1600 (B) 1618 (C) 1627 (D) 1609

Q44. JEE Main 2024 (Online) 8th April Evening Shift

Let $\overrightarrow{a}=\hat{i}+2\hat{j}+3\hat{k},\overrightarrow{b}=2\hat{i}+3\hat{j}-5\hat{k}$ and $\overrightarrow{c}=3\hat{i}-\hat{j}+\lambda\hat{k}$ be three vectors. Let $\overrightarrow{r}$ be a unit vector along $\vec{b}+\vec{c}$. If $\vec{r}\cdot\vec{a}=3$, then 3λ is equal to:

(A) 21 (B) 25 (C) 27 (D) 30

Q45. JEE Main 2024 (Online) 8th April Morning Shift

The set of all α, for which the vectors $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$, is

(A) $[0, 1)$ (B) $\left(-\frac{4}{3}, 0\right]$ (C) $(-2, 0]$ (D) $\left(-\frac{4}{3}, 1\right)$

Q46. JEE Main 2024 (Online) 6th April Evening Shift

Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}, \vec{b} = ((\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i}) \times \hat{i}$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is:

(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) 2 (D) $\frac{2}{3}$

Q47. JEE Main 2024 (Online) 6th April Evening Shift

Let $\overrightarrow{a} = 6\hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{b} = \hat{i} + \hat{j}$. If $\overrightarrow{c}$ is a vector such that $|\overrightarrow{c}| \geq 6, \overrightarrow{a} \cdot \overrightarrow{c} = 6|\overrightarrow{c}|, |\overrightarrow{c} - \overrightarrow{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 60°, then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to:

(A) $\frac{3}{2}\sqrt{6}$ (B) $\frac{9}{2}(6 - \sqrt{6})$ (C) $\frac{9}{2}(6 + \sqrt{6})$ (D) $\frac{3}{2}\sqrt{3}$

Q48. JEE Main 2024 (Online) 6th April Morning Shift

If $A(3, 1, -1), B\left(\frac{5}{3}, \frac{7}{3}, \frac{1}{3}\right), C(2, 2, 1)$ and $D\left(\frac{10}{3}, \frac{2}{3}, \frac{-1}{3}\right)$ are the vertices of a quadrilateral $ABCD$, then its area is

(A) $\frac{4\sqrt{2}}{3}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $\frac{5\sqrt{2}}{3}$ (D) $2\sqrt{2}$

Q49. JEE Main 2024 (Online) 5th April Evening Shift

Let $\vec{a} = 2\hat{i} + 5\hat{j} - \hat{k}, \vec{b} = 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{c}$ be three vectors such that $(\vec{c} + \hat{i}) \times (\vec{a} + \vec{b} + \hat{i}) = \vec{a} \times (\vec{c} + \hat{i})$. If $\vec{a} \cdot \vec{c} = -29$, then $\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k})$ is equal to:

(A) 15 (B) 10 (C) 5 (D) 12

Q50. JEE Main 2024 (Online) 5th April Evening Shift

Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in \left[0, \frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to:

(A) 124 (B) 110 (C) 121 (D) 105

Q51. JEE Main 2024 (Online) 5th April Morning Shift

If $\mathrm{A}(1, -1, 2), \mathrm{B}(5, 7, -6), \mathrm{C}(3, 4, -10)$ and $\mathrm{D}(-1, -4, -2)$ are the vertices of a quadrilateral ABCD, then its area is :

(A) $24\sqrt{7}$ (B) $48\sqrt{7}$ (C) $24\sqrt{29}$ (D) $12\sqrt{29}$

Q52. JEE Main 2024 (Online) 4th April Evening Shift

For $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are mutually perpendicular, then the value of $(14\cos\theta)^2$ is equal to

(A) 25 (B) 50 (C) 20 (D) 40

Q53. JEE Main 2024 (Online) 4th April Evening Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}, x \in \mathbb{R}$. If $\vec{d}$ is the unit vector in the direction of $\vec{b} + \vec{c}$ such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to

(A) 3 (B) 9 (C) 11 (D) 6

Q54. JEE Main 2024 (Online) 4th April Morning Shift

Let a unit vector which makes an angle of 60° with $2\hat{i} + 2\hat{j} - \hat{k}$ and an angle of 45° with $\hat{i} - \hat{k}$ be $\vec{C}$. Then $\vec{C} + \left(-\frac{1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$ is:

(A) $-\frac{\sqrt{2}}{3}\hat{i} + \frac{\sqrt{2}}{3}\hat{j} + \left(\frac{1}{2} + \frac{2\sqrt{2}}{3}\right)\hat{k}$

(B) $\left(\frac{1}{\sqrt{3}} + \frac{1}{2}\right)\hat{i} + \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{2}}\right)\hat{j} + \left(\frac{1}{\sqrt{3}} + \frac{\sqrt{2}}{3}\right)\hat{k}$

(C) $\frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$

(D) $\frac{\sqrt{2}}{3}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{1}{2}\hat{k}$

Q55. JEE Main 2024 (Online) 1st February Evening Shift

Consider a $\triangle ABC$ where $A(1, 3, 2), B(-2, 8, 0)$ and $C(3, 6, 7)$. If the angle bisector of $\angle BAC$ meets the line BC at D, then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is :

(A) $\frac{37}{2\sqrt{38}}$ (B) $\sqrt{19}$ (C) $\frac{39}{2\sqrt{38}}$ (D) $\frac{\sqrt{38}}{2}$

Q56. JEE Main 2024 (Online) 1st February Morning Shift

Let $\overrightarrow{a} = -5\hat{i} + \hat{j} - 3\hat{k}, \overrightarrow{b} = \hat{i} + 2\hat{j} - 4\hat{k}$ and

$\overrightarrow{c} = (((\overrightarrow{a} \times \overrightarrow{b}) \times \hat{i}) \times \hat{i}) \times \hat{i}$. Then $\vec{c} \cdot (-\hat{i} + \hat{j} + \hat{k})$ is equal to :

(A) -12 (B) -10 (C) -13 (D) -15

Q57. JEE Main 2024 (Online) 31st January Morning Shift

Let $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = 4\hat{i} + \hat{j} + 7\hat{k}$ and $\vec{c} = \hat{i} - 3\hat{j} + 4\hat{k}$ be three vectors. If a vector $\vec{p}$ satisfies $\vec{p} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{p} \cdot \vec{a} = 0$, then $\vec{p} \cdot (\hat{i} - \hat{j} - \hat{k})$ is equal to

(A) 24 (B) 32 (C) 36 (D) 28

Q58. JEE Main 2024 (Online) 30th January Evening Shift

Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}, \alpha, \beta \in \mathbb{R}$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $|\vec{b}|^2 = 6$. If $\vec{a} \cdot \vec{b} = 3\sqrt{2}$, then the value of $(\alpha^2 + \beta^2)|\vec{a} \times \vec{b}|^2$ is equal to

(A) 85 (B) 90 (C) 75 (D) 95

Q59. JEE Main 2024 (Online) 30th January Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{b}| = 1$ and $|\vec{b} \times \vec{a}| = 2$. Then $|(\vec{b} \times \vec{a}) - \vec{b}|^2$ is equal to

(A) 1 (B) 3 (C) 5 (D) 4

Q60. JEE Main 2024 (Online) 30th January Morning Shift

Let $\overrightarrow{\mathrm{a}} = \mathrm{a}_1\hat{i} + \mathrm{a}_2\hat{j} + \mathrm{a}_3\hat{k}$ and $\overrightarrow{\mathrm{b}} = \mathrm{b}_1\hat{i} + \mathrm{b}_2\hat{j} + \mathrm{b}_3\hat{k}$ be two vectors such that $|\overrightarrow{\mathrm{a}}| = 1, \vec{a} \cdot \vec{b} = 2$ and $|\vec{b}| = 4$. If $\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to:

(A) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$

(C) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

Q61. JEE Main 2024 (Online) 30th January Morning Shift

Let $A(2, 3, 5)$ and $C(-3, 4, -2)$ be opposite vertices of a parallelogram $ABCD$. If the diagonal $\overrightarrow{\mathrm{BD}} = \hat{i} + 2\hat{j} + 3\hat{k}$, then the area of the parallelogram is equal to :

(A) $\frac{1}{2}\sqrt{410}$ (B) $\frac{1}{2}\sqrt{306}$ (C) $\frac{1}{2}\sqrt{586}$ (D) $\frac{1}{2}\sqrt{474}$

Q62. JEE Main 2024 (Online) 29th January Evening Shift

Let a unit vector $\hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}, \frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}, \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ then $|\hat{u} - \vec{v}|^2$ is equal to

(A) $\frac{11}{2}$ (B) $\frac{5}{2}$ (C) 7 (D) 9

Q63. JEE Main 2024 (Online) 29th January Evening Shift

Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 12\vec{a} + 4\vec{b}$ and $\overrightarrow{OC} = \vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then $\frac{\text{area of the quadrilateral OA BC}}{\text{area of S}}$ is equal to _____.

(A) 7 (B) 6 (C) 8 (D) 10

Q64. JEE Main 2024 (Online) 29th January Morning Shift

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear. If $\vec{a} + 5\vec{b}$ is collinear with $\vec{c}, \vec{b} + 6\vec{c}$ is collinear with $\vec{a}$ and $\vec{a} + \alpha\vec{b} + \beta\vec{c} = \overrightarrow{0}$, then $\alpha + \beta$ is equal to

(A) 30 (B) -30 (C) -25 (D) 35

Q65. JEE Main 2024 (Online) 29th January Morning Shift

Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of $\angle \mathrm{AOB}$ meets the line AB at C, then the length of OC is

(A) $\frac{3}{2}\sqrt{34}$ (B) $\frac{2}{3}\sqrt{31}$ (C) $\frac{2}{3}\sqrt{34}$ (D) $\frac{3}{2}\sqrt{31}$

Q66. JEE Main 2024 (Online) 27th January Evening Shift

Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i}+2\hat{j}+\hat{k}, \hat{i}+2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$ respectively. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the orthocentre of the triangle on the sides AB, BC and CA respectively, then $l_1^2+l_2^2+l_3^2$ equals:

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Q67. JEE Main 2024 (Online) 27th January Evening Shift

The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}, 2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let l denote the length of the angle bisector AD of $\angle \mathrm{BAC}$ where D is on the line segment BC, then $2l^2$ equals :

(A) 45 (B) 50 (C) 42 (D) 49

Q68. JEE Main 2024 (Online) 27th January Morning Shift

Let $\overrightarrow{\mathrm{a}}=\hat{i}+2\hat{j}+\hat{k}$,
$\overrightarrow{\mathrm{b}}=3(\hat{i}-\hat{j}+\hat{k})$.
Let $\overrightarrow{c}$ be the vector such that $\overrightarrow{\mathrm{a}}\times\overrightarrow{c}=\overrightarrow{\mathrm{b}}$ and $\vec{a}\cdot\vec{c}=3$.
Then $\vec{a}\cdot((\vec{c}\times\vec{b})-\vec{b}-\vec{c})$ is equal to :

(A) 32 (B) 36 (C) 24 (D) 20

Q69. JEE Main 2023 (Online) 15th April Morning Shift

Let ABCD be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\overrightarrow{AB}-\overrightarrow{BC})+(\overrightarrow{AD}-\overrightarrow{DC})=k\overrightarrow{FE}$, then k is equal to :

(A) -2 (B) 4 (C) -4 (D) 2

Q70. JEE Main 2023 (Online) 13th April Evening Shift

Let $|\vec{a}|=2, |\vec{b}|=3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $|(\vec{a}+2\vec{b})\times(2\vec{a}-3\vec{b})|^2$ is equal to :

(A) 441 (B) 482 (C) 841 (D) 882

Q71. JEE Main 2023 (Online) 13th April Evening Shift

Let for a triangle ABC,

$\overrightarrow{\mathrm{AB}} = -2\hat{i} + \hat{j} + 3\hat{k}$

$\overrightarrow{\mathrm{CB}} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\overrightarrow{\mathrm{CA}} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$

If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$, then $\overrightarrow{CB} \cdot \overrightarrow{CA}$ is equal to

(A) 60 (B) 54 (C) 120 (D) 108

Q72. JEE Main 2023 (Online) 13th April Morning Shift

Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 24$, then $|\vec{d}|^2$ is equal to :

(A) 313 (B) 413 (C) 423 (D) 323

Q73. JEE Main 2023 (Online) 11th April Morning Shift

For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10\,|a_i| < 1, i = 1, 2, 3$, consider the following statements :

(A): $\max\{|a_1|, |a_2|, |a_3|\} \leq |\vec{a}|$

(B) : $|\vec{a}| \leq 3\max\{|a_1|, |a_2|, |a_3|\}$

(A) Only (B) is true (B) Only (A) is true

(C) Neither (A) nor (B) is true (D) Both (A) and (B) are true

Q74. JEE Main 2023 (Online) 11th April Morning Shift

Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to :

(A) $\left(\frac{\pi}{3}, 3\sqrt{6}\right)$ (B) $\left(\frac{\pi}{3}, 6\right)$

(C) $\left(\frac{\pi}{4}, 3\sqrt{6}\right)$ (D) $\left(\frac{\pi}{4}, 6\right)$

Q75. JEE Main 2023 (Online) 10th April Evening Shift

Let $\vec{a} = 2\hat{i} + 7\hat{j} - \hat{k}, \vec{b} = 3\hat{i} + 5\hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 12$. Then $(-\hat{i} + \hat{j} - \hat{k}) \cdot (\vec{c} \times \vec{d})$ is equal to :

(A) 24 (B) 42 (C) 44 (D) 48

Q76. JEE Main 2023 (Online) 10th April Evening Shift

If the points P and Q are respectively the circumcenter and the orthocentre of a $\triangle \mathrm{ABC}$, then $\overrightarrow{\mathrm{PA}}+\overrightarrow{\mathrm{PB}}+\overrightarrow{\mathrm{PC}}$ is equal to :

(A) $\overrightarrow{QP}$ (B) $\overrightarrow{PQ}$ (C) $2\overrightarrow{PQ}$ (D) $2\overrightarrow{QP}$

Q77. JEE Main 2023 (Online) 10th April Morning Shift

Let O be the origin and the position vector of the point P be $-\hat{i}-2\hat{j}+3\hat{k}$. If the position vectors of the points A, B and C are $-2\hat{i}+\hat{j}-3\hat{k}, 2\hat{i}+4\hat{j}-2\hat{k}$ and $-4\hat{i}+2\hat{j}-\hat{k}$ respectively, then the projection of the vector $\overrightarrow{OP}$ on a vector perpendicular to the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is :

(A) $\frac{7}{3}$ (B) 3 (C) $\frac{10}{3}$ (D) $\frac{8}{3}$

Q78. JEE Main 2023 (Online) 10th April Morning Shift

An arc PQ of a circle subtends a right angle at its centre O. The mid point of the arc PQ is R. If $\overrightarrow{OP}=\vec{u}, \overrightarrow{OR}=\vec{v}$, and $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$, then α, β^2 are the roots of the equation :

(A) $x^2+x-2=0$ (B) $3x^2+2x-1=0$ (C) $3x^2-2x-1=0$ (D) $x^2-x-2=0$

Q79. JEE Main 2023 (Online) 8th April Evening Shift

The area of the quadrilateral ABCD with vertices $\mathrm{A}(2,1,1), \mathrm{B}(1,2,5), \mathrm{C}(-2,-3,5)$ and $\mathrm{D}(1,-6,-7)$ is equal to :

(A) 48 (B) $8\sqrt{38}$ (C) 54 (D) $9\sqrt{38}$

Q80. JEE Main 2023 (Online) 8th April Morning Shift

If the points with position vectors $\alpha\hat{i}+10\hat{j}+13\hat{k}, 6\hat{i}+11\hat{j}+11\hat{k}, \frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ are collinear, then $(19\alpha-6\beta)^2$ is equal to :

(A) 16 (B) 49 (C) 36 (D) 25

Q81. JEE Main 2023 (Online) 6th April Morning Shift

Let $\vec{a}=2\hat{i}+3\hat{j}+4\hat{k}, \vec{b}=\hat{i}-2\hat{j}-2\hat{k}$ and $\vec{c}=-\hat{i}+4\hat{j}+3\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$, and $\vec{a}\cdot\vec{d}=18$, then $|\vec{a}\times\vec{d}|^2$ is equal to :

(A) 680 (B) 720 (C) 760 (D) 640

Q82. JEE Main 2023 (Online) 1st February Evening Shift

Let $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + 5\hat{k}$ be two vectors. Then which one of the following statements is TRUE ?

(A) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(B) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(C) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

(D) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

Q83. JEE Main 2023 (Online) 1st February Evening Shift

Let $\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}, \vec{b} = \hat{i} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $|\vec{r}|$ is equal to :

(A) $\frac{11}{7}$ (B) $\frac{11}{5}\sqrt{2}$ (C) $\frac{\sqrt{914}}{7}$ (D) $\frac{11}{7}\sqrt{2}$

Q84. JEE Main 2023 (Online) 31st January Evening Shift

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$ be three vectors. If $\vec{r}$ is a vector such that, $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then $25|\vec{r}|^2$ is equal to :

(A) 336 (B) 449 (C) 339 (D) 560

Q85. JEE Main 2023 (Online) 31st January Morning Shift

Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, and $\vec{b}$ and $\vec{c}$ be two nonzero vectors such that $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$ and $\vec{b} \cdot \vec{c} = 0$. Consider the following two statements:

(A) $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$ for all $\lambda \in \mathbb{R}$.

(B) $\vec{a}$ and $\vec{c}$ are always parallel.

Then,

(A) only (B) is correct (B) both (A) and (B) are correct

(C) only (A) is correct (D) neither (A) nor (B) is correct

Q86. JEE Main 2023 (Online) 30th January Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors, Let $|\vec{a}| = 1, |\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$, then the value of $\vec{b} \cdot \vec{c}$ is :

(A) -48 (B) -60 (C) -84 (D) -24

Q87. JEE Main 2023 (Online) 30th January Morning Shift

Let a unit vector $\widehat{OP}$ make angles α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$. If $\widehat{\mathrm{OP}}$ is perpendicular to the plane through points $(1,2,3),(2,3,4)$ and $(1,5,7)$, then which one of the following is true?

(A) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(B) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(C) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(D) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

Q88. JEE Main 2023 (Online) 29th January Evening Shift

If $\vec{a} = \hat{i} + 2\hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}, \vec{c} = 7\hat{i} - 3\hat{j} + 4\hat{k}, \vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$. Then $\vec{r} \cdot \vec{c}$ is equal to :

(A) 36 (B) 30 (C) 34 (D) 32

Q89. JEE Main 2023 (Online) 29th January Evening Shift

Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$. If $\vec{c}$ is a vector such that $\vec{c} \cdot \left(\vec{a} \times \vec{b}\right) + 25 = 0, \vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals :

(A) $\frac{3}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{5}$ (D) $\frac{5}{\sqrt{2}}$

Q90. JEE Main 2023 (Online) 29th January Morning Shift

If the vectors $\vec{a} = \lambda\hat{i} + \mu\hat{j} + 4\hat{k}, \vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$ are coplanar and the projection of $\vec{a}$ on the vector $\vec{b}$ is $\sqrt{54}$ units, then the sum of all possible values of $\lambda + \mu$ is equal to :

(A) 24 (B) 0 (C) 18 (D) 6

Q91. JEE Main 2023 (Online) 25th January Morning Shift

The vector $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a} + \sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ is :

(A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) 1 (D) $3\sqrt{2}$

Q92. JEE Main 2023 (Online) 24th January Evening Shift

Let $\vec{\alpha} = 4\hat{i} + 3\hat{j} + 5\hat{k}$ and $\vec{\beta} = \hat{i} + 2\hat{j} - 4\hat{k}$. Let $\vec{\beta}_1$ be parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta} = \vec{\beta}_1 + \vec{\beta}_2$, then the value of $5\vec{\beta}_2 \cdot \left(\hat{i} + \hat{j} + \hat{k}\right)$ is :

(A) 9 (B) 7 (C) 6 (D) 11

Q93. JEE Main 2023 (Online) 24th January Morning Shift

Let PQR be a triangle. The points A, B and C are on the sides QR, RP and PQ respectively such that

$\frac{QA}{AR} = \frac{RB}{BP} = \frac{PC}{CQ} = \frac{1}{2}$. Then $\frac{Area(\Delta PQR)}{Area(\Delta ABC)}$ is equal to :

(A) $\frac{5}{2}$ (B) 4 (C) 2 (D) 3

Q94. JEE Main 2022 (Online) 29th July Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar concurrent vectors such that the angle between any two of them is the same. If the product of their magnitudes is 14 and
$(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = 168$, then $|\vec{a}| + |\vec{b}| + |\vec{c}|$ is equal to :

(A) 10 (B) 14 (C) 16 (D) 18

Q95. JEE Main 2022 (Online) 29th July Morning Shift

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a} + \hat{b})$ and $(\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b}))$, then the value of $164 \cos^2\theta$ is equal to :

(A) $90 + 27\sqrt{2}$ (B) $45 + 18\sqrt{2}$ (C) $90 + 3\sqrt{2}$ (D) $54 + 90\sqrt{2}$

Q96. JEE Main 2022 (Online) 28th July Evening Shift

Let S be the set of all $a \in R$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}$, $(b > 1)$ is acute. Then S is equal to :

(A) $(-\infty, -\frac{4}{3})$ (B) Φ
(C) $(-\frac{4}{3}, 0)$ (D) $(\frac{12}{7}, \infty)$

Q97. JEE Main 2022 (Online) 28th July Morning Shift

Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}, \vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}, t \in \mathbf{R}$ be such that for $\alpha, \beta, \gamma \in \mathbf{R}, \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \Rightarrow \alpha = \beta = \gamma = 0$. Then, the set of all values of t is :

(A) a non-empty finite set (B) equal to $\mathbf{N}$
(C) equal to $\mathbf{R} - \{0\}$ (D) equal to $\mathbf{R}$

Q98. JEE Main 2022 (Online) 28th July Morning Shift

Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbf{R} \times \mathbf{R} - \{(0, 0)\}$, the vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$. Then the value of $|\vec{a} \times \vec{b}|$ is equal to :

(A) $9\sqrt{3}$ (B) $27\sqrt{3}$ (C) 9 (D) 81

Q99. JEE Main 2022 (Online) 27th July Morning Shift

Let $\vec{a} = \alpha\hat{i} + \hat{j} + \beta\hat{k}$ and $\vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$ be two vectors, such that $\vec{a} \times \vec{b} = -\hat{i} + 9\hat{j} + 12\hat{k}$. Then the projection of $\vec{b} - 2\vec{a}$ on $\vec{b} + \vec{a}$ is equal to :

(A) 2 (B) $\frac{39}{5}$ (C) 9 (D) $\frac{46}{5}$

Q100. JEE Main 2022 (Online) 26th July Morning Shift

Let $\overrightarrow{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}, \alpha > 0$. If the projection of $\overrightarrow{a} \times \overrightarrow{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$ is 30, then α is equal to :

(A) $\frac{15}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 7

Q101. JEE Main 2022 (Online) 25th July Evening Shift

Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a} \times \vec{b} = 2\hat{i} - \hat{k}$ and $\vec{a} \cdot \vec{b} = 3$. Then the projection of $\vec{b}$ on the vector $\vec{a} - \vec{b}$ is :

(A) $\frac{2}{\sqrt{21}}$ (B) $2\sqrt{\frac{3}{7}}$ (C) $\frac{2}{3}\sqrt{\frac{7}{3}}$ (D) $\frac{2}{3}$

Q102. JEE Main 2022 (Online) 25th July Morning Shift

Let ABC be a triangle such that $\overrightarrow{BC} = \overrightarrow{a}, \overrightarrow{CA} = \overrightarrow{b}, \overrightarrow{AB} = \overrightarrow{c}, |\overrightarrow{a}| = 6\sqrt{2}, |\overrightarrow{b}| = 2\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 12$. Consider the statements :

$(S1) : |(\overrightarrow{a} \times \overrightarrow{b}) + (\overrightarrow{c} \times \overrightarrow{b})| - |\vec{c}| = 6(2\sqrt{2} - 1)$

$(S2) : \angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then

(A) both (S1) and (S2) are true (B) only (S1) is true

(C) only (S2) is true (D) both (S1) and (S2) are false

Q103. JEE Main 2022 (Online) 29th June Evening Shift

Let A, B, C be three points whose position vectors respectively are

$\overrightarrow{a} = \hat{i} + 4\hat{j} + 3\widehat{k}$

$\overrightarrow{b} = 2\hat{i} + \alpha\hat{j} + 4\widehat{k}, \alpha \in R$

$\overrightarrow{c} = 3\hat{i} - 2\hat{j} + 5\widehat{k}$

If α is the smallest positive integer for which $\overrightarrow{a}, \overrightarrow{b}, \overrightarrow{c}$ are noncollinear, then the length of the median, in ΔABC, through A is :

(A) $\frac{\sqrt{82}}{2}$ (B) $\frac{\sqrt{62}}{2}$ (C) $\frac{\sqrt{69}}{2}$ (D) $\frac{\sqrt{66}}{2}$

Q104. JEE Main 2022 (Online) 29th June Morning Shift

Let $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}, \vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ where $\alpha, \beta \in R$, be three vectors. If the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$, then the value of $\alpha+\beta$ is equal to :

(A) 3 (B) 4 (C) 5 (D) 6

Q105. JEE Main 2022 (Online) 28th June Evening Shift

Let $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$, where $\alpha \in R$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15(\alpha^2+4)}$, then the value of $2\left|\vec{a}\right|^2+\left(\vec{a}.\vec{b}\right)\left|\vec{b}\right|^2$ is equal to :

(A) 10 (B) 7 (C) 9 (D) 14

Q106. JEE Main 2022 (Online) 28th June Evening Shift

Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$. If $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i}+2\hat{j}+\hat{k}$ is :

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{5}{3}$ (D) $\frac{7}{3}$

Q107. JEE Main 2022 (Online) 27th June Evening Shift

Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonals of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute, $|\vec{a}|=1$, and $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$. If $\vec{c}=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is :

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{6}$ (D) $\frac{3\pi}{4}$

Q108. JEE Main 2022 (Online) 27th June Morning Shift

Let $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$. Then the number of vectors $\vec{b}$ such that $\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{b}| \in \{1, 2, ..., 10\}$ is :

(A) 0 (B) 1 (C) 2 (D) 3

Q109. JEE Main 2022 (Online) 26th June Evening Shift

Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k}, \vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v}.\hat{j}=7$, then $\vec{v}.\left(\hat{i}+\hat{k}\right)$ is equal to :

(A) 6 (B) 7 (C) 8 (D) 9

Q110. JEE Main 2022 (Online) 25th June Morning Shift

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $a_i > 0, i = 1, 2, 3$ be a vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a}, \vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to:

(A) $\sqrt{7}$ (B) $\sqrt{2}$ (C) 2 (D) 7

Q111. JEE Main 2022 (Online) 24th June Evening Shift

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : The projection of $\hat{a}$ on $(\hat{a} + \hat{b})$ is $\frac{1}{2}$

(A) Only (S1) is true. (B) Only (S2) is true.

(C) Both (S1) and (S2) are true. (D) Both (S1) and (S2) are false.

Q112. JEE Main 2022 (Online) 24th June Morning Shift

Let $\hat{a}, \hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\vec{c}$ is $\frac{\pi}{12}$, and $\hat{b} = \vec{c} + 2\left(\vec{c} \times \hat{a}\right)$, then $\left|6\vec{c}\right|^2$ is equal to :

(A) $6\left(3 - \sqrt{3}\right)$ (B) $3 + \sqrt{3}$ (C) $6\left(3 + \sqrt{3}\right)$ (D) $6\left(\sqrt{3} + 1\right)$

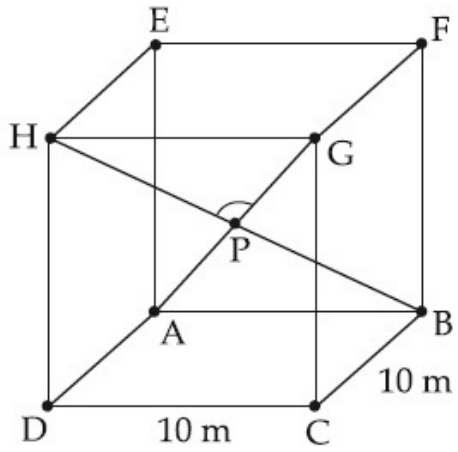
Q113. JEE Main 2021 (Online) 31st August Morning Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|2\vec{a} + 3\vec{b}\right| = \left|3\vec{a} + \vec{b}\right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60°. If $\frac{1}{8}\vec{a}$ is a unit vector, then $\left|\vec{b}\right|$ is equal to :

(A) 4 (B) 6 (C) 5 (D) 8

Q114. JEE Main 2021 (Online) 26th August Evening Shift

A hall has a square floor of dimension 10 m $\times$ 10 m (see the figure) and vertical walls. If the angle GPH between the diagonals AG and BH is $\cos^{-1}\frac{1}{5}$, then the height of the hall (in meters) is :



(A) 5 (B) $2\sqrt{10}$ (C) $5\sqrt{3}$ (D) $5\sqrt{2}$

Q115. JEE Main 2021 (Online) 27th July Morning Shift

Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+2\hat{j}+3\hat{k}$. Then the vector product $\left(\vec{a}+\vec{b}\right)\times\left(\left(\vec{a}\times\left(\left(\vec{a}-\vec{b}\right)\times\vec{b}\right)\right)\times\vec{b}\right)$ is equal to :

(A) $5(34\hat{i}-5\hat{j}+3\hat{k})$ (B) $7(34\hat{i}-5\hat{j}+3\hat{k})$

(C) $7(30\hat{i}-5\hat{j}+7\hat{k})$ (D) $5(30\hat{i}-5\hat{j}+7\hat{k})$

Q116. JEE Main 2021 (Online) 25th July Evening Shift

Let a, b and c be distinct positive numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are co-planar, then c is equal to :

(A) $\frac{2}{\frac{1}{a}+\frac{1}{b}}$ (B) $\frac{a+b}{2}$

(C) $\frac{1}{a}+\frac{1}{b}$ (D) $\sqrt{ab}$

Q117. JEE Main 2021 (Online) 25th July Evening Shift

If $\left|\vec{a}\right|=2, \left|\vec{b}\right|=5$ and $\left|\vec{a}\times\vec{b}\right|$ = 8, then $\left|\vec{a}.\vec{b}\right|$ is equal to :

(A) 6 (B) 4 (C) 3 (D) 5

Q118. JEE Main 2021 (Online) 20th July Evening Shift

In a triangle ABC, if $\left|\overrightarrow{BC}\right| = 3$, $\left|\overrightarrow{CA}\right| = 5$ and $\left|\overrightarrow{BA}\right| = 7$, then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to :

(A) $\frac{19}{2}$ (B) $\frac{13}{2}$ (C) $\frac{11}{2}$ (D) $\frac{15}{2}$

Q119. JEE Main 2021 (Online) 18th March Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between the vectors $\left(\vec{a} + \vec{b} + \left(\vec{a} \times \vec{b}\right)\right)$ and $\vec{a}$ is equal to :

(A) $\sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (D) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Q120. JEE Main 2021 (Online) 18th March Evening Shift

In a triangle ABC, if $|\overrightarrow{BC}| = 8, |\overrightarrow{CA}| = 7, |\overrightarrow{AB}| = 10$, then the projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to :

(A) $\frac{25}{4}$ (B) $\frac{127}{20}$ (C) $\frac{85}{14}$ (D) $\frac{115}{16}$

Q121. JEE Main 2021 (Online) 18th March Morning Shift

A vector $\vec{a}$ has components 3p and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components p + 1 and $\sqrt{10}$, then the value of p is equal to :

(A) 1 (B) $-\frac{5}{4}$ (C) $\frac{4}{5}$ (D) -1

Q122. JEE Main 2021 (Online) 17th March Evening Shift

Let O be the origin. Let $\overrightarrow{OP} = x\hat{i} + y\hat{j} - \hat{k}$ and $\overrightarrow{OQ} = -\hat{i} + 2\hat{j} + 3x\hat{k}$, x, y$\in$R, x > 0, be such that $\left|\overrightarrow{PQ}\right| = \sqrt{20}$ and the vector $\overrightarrow{OP}$ is perpendicular $\overrightarrow{OQ}$. If $\overrightarrow{OR} = 3\hat{i} + z\hat{j} - 7\hat{k}$, z$\in$R, is coplanar with $\overrightarrow{OP}$ and $\overrightarrow{OQ}$, then the value of $x^2 + y^2 + z^2$ is equal to :

(A) 2 (B) 9 (C) 7 (D) 1

Q123. JEE Main 2021 (Online) 16th March Evening Shift

Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$,

$\vec{r} \cdot \left(\alpha\hat{i} + 2\hat{j} + \hat{k}\right) = 3$ and $\vec{r} \cdot \left(2\hat{i} + 5\hat{j} - \alpha\hat{k}\right) = -1$, $\alpha \in R$, then the

value of $\alpha + \left|\vec{r}\right|^2$ is equal to :

(A) 13 (B) 11 (C) 9 (D) 15

Q124. JEE Main 2021 (Online) 16th March Morning Shift

Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counterclockwise direction in the first quadrant. Then the area of triangle having vertices (α, β), (0, β) and (0, 0) is equal to :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 1 (D) $2\sqrt{2}$

Q125. JEE Main 2021 (Online) 26th February Evening Shift

If vectors $\vec{a_1} = x\hat{i} - \hat{j} + \hat{k}$ and $\vec{a_2} = \hat{i} + y\hat{j} + z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i} + y\hat{j} + z\hat{k}$ is :

(A) $\frac{1}{\sqrt{3}}\left(\hat{i} - \hat{j} + \hat{k}\right)$ (B) $\frac{1}{\sqrt{2}}\left(-\hat{j} + \hat{k}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\hat{i} - \hat{j}\right)$ (D) $\frac{1}{\sqrt{3}}\left(\hat{i} + \hat{j} - \hat{k}\right)$

Q126. JEE Main 2020 (Online) 3rd September Evening Slot

Let a, b, c $\in$ R be such that $a^2 + b^2 + c^2 = 1$. If
$a\cos\theta = b\cos\left(\theta + \frac{2\pi}{3}\right) = c\cos\left(\theta + \frac{4\pi}{3}\right)$,
where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is :

(A) 0 (B) $\frac{\pi}{9}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{2}$

Q127. JEE Main 2020 (Online) 8th January Evening Slot

Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$, then $\vec{c} \cdot \vec{b}$ is equal to

(A) $-\frac{1}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{1}{2}$ (D) -1

Q128. JEE Main 2020 (Online) 7th January Evening Slot

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. If $\lambda=\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ and $\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$, then the ordered pair, $\left(\lambda,\vec{d}\right)$ is equal to :

(A) $\left(\frac{3}{2},3\vec{a}\times\vec{c}\right)$

(B) $\left(-\frac{3}{2},3\vec{c}\times\vec{b}\right)$

(C) $\left(-\frac{3}{2},3\vec{a}\times\vec{b}\right)$

(D) $\left(\frac{3}{2},3\vec{b}\times\vec{c}\right)$

Q129. JEE Main 2020 (Online) 7th January Morning Slot

A vector $\vec{a}=\alpha\hat{i}+2\hat{j}+\beta\hat{k}\,(\alpha,\beta\in R)$ lies in the plane of the vectors, $\vec{b}=\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}-\hat{j}+4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then:

(A) $\vec{a}.\hat{i}+3=0$

(B) $\vec{a}.\hat{k}-4=0$

(C) $\vec{a}.\hat{i}+1=0$

(D) $\vec{a}.\hat{k}+2=0$

Q130. JEE Main 2019 (Online) 12th April Morning Slot

Let $\vec{a}=3\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-2\hat{k}$ be two vectors. If a vector perpendicular to both the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ has the magnitude 12 then one such vector is :

(A) $4\left(2\hat{i}-2\hat{j}-\hat{k}\right)$

(B) $4\left(-2\hat{i}-2\hat{j}+\hat{k}\right)$

(C) $4\left(2\hat{i}+2\hat{j}+\hat{k}\right)$

(D) $4\left(2\hat{i}+2\hat{j}-\hat{k}\right)$

Q131. JEE Main 2019 (Online) 10th April Morning Slot

Let A (3, 0, −1), B(2, 10, 6) and C(1, 2, 1) be the vertices of a triangle and M be the midpoint of AC. If G divides BM in the ratio, 2 : 1, then cos (∠GOA) (O being the origin) is equal to :

(A) $\frac{1}{\sqrt{15}}$

(B) $\frac{1}{6\sqrt{10}}$

(C) $\frac{1}{\sqrt{30}}$

(D) $\frac{1}{2\sqrt{15}}$

Q132. JEE Main 2019 (Online) 9th April Evening Slot

If a unit vector $\vec{a}$ makes angles $\pi/3$ with $\hat{i}$, $\pi/4$ with $\hat{j}$ and $\theta\in(0,\pi)$ with $\hat{k}$, then a value of θ is :-

(A) $\frac{5\pi}{6}$

(B) $\frac{5\pi}{12}$

(C) $\frac{2\pi}{3}$

(D) $\frac{\pi}{4}$

Q133. JEE Main 2019 (Online) 9th April Morning Slot

Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. If $\vec{\beta} = \vec{\beta_1} - \vec{\beta_2}$, where $\vec{\beta_1}$ is parallel to $\vec{\alpha}$ and $\vec{\beta_2}$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta_1} \times \vec{\beta_2}$ is equal to

(A) $3\hat{i} - 9\hat{j} - 5\hat{k}$

(B) $\frac{1}{2}(-3\hat{i} + 9\hat{j} + 5\hat{k})$

(C) $-3\hat{i} + 9\hat{j} + 5\hat{k}$

(D) $\frac{1}{2}(3\hat{i} - 9\hat{j} + 5\hat{k})$

Q134. JEE Main 2019 (Online) 8th April Evening Slot

Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x. Then $\left|\vec{a} \times \vec{b}\right|$ = r is possible if :

(A) $0 < r < \sqrt{\frac{3}{2}}$

(B) $3\sqrt{\frac{3}{2}} < r < 5\sqrt{\frac{3}{2}}$

(C) $r \geq 5\sqrt{\frac{3}{2}}$

(D) $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$

Q135. JEE Main 2019 (Online) 11th January Evening Slot

Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1-\beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all possible values of β is :

(A) 4

(B) 1

(C) 2

(D) 3

Q136. JEE Main 2019 (Online) 11th January Morning Slot

Let $\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k}, \vec{b} = \hat{i} + \lambda\hat{j} + 4\hat{k}$ and $\vec{c} = 2\hat{i} + 4\hat{j} + (\lambda^2 - 1)\hat{k}$ be coplanar vectors. Then the non-zero vector $\vec{a} \times \vec{c}$ is :

(A) $-10\hat{i} - 5\hat{j}$

(B) $-10\hat{i} + 5\hat{j}$

(C) $-14\hat{i} + 5\hat{j}$

(D) $-14\hat{i} - 5\hat{j}$

Q137. JEE Main 2019 (Online) 10th January Evening Slot

If $\vec{\alpha} = (\lambda - 2)\vec{a} + \vec{b}$ and $\vec{\beta} = (4\lambda - 2)\vec{a} + 3\vec{b}$ be two given vectors $\vec{a}$ and $\vec{b}$ are non-collinear. The value of λ for which vectors $\vec{\alpha}$ and $\vec{\beta}$ are collinear, is -

(A) 4

(B) 3

(C) -3

(D) -4

Q138. JEE Main 2019 (Online) 10th January Morning Slot

Let $\vec{a} = 2\hat{i} + \lambda_1\hat{j} + 3\hat{k}$, $\vec{b} = 4\hat{i} + (3 - \lambda_2)\hat{j} + 6\hat{k}$, and $\vec{c} = 3\hat{i} + 6\hat{j} + (\lambda_3 - 1)\hat{k}$ be three vectors such that $\vec{b} = 2\vec{a}$ and $\vec{a}$ is perpendicular to $\vec{c}$. Then a possible value of $(\lambda_1, \lambda_2, \lambda_3)$ is :

(A) (1, 5, 1)

(B) (1, 3, 1)

(C) $\left(-\frac{1}{2}, 4, 0\right)$

(D) $\left(\frac{1}{2}, 4, -2\right)$

Q139. JEE Main 2019 (Online) 9th January Evening Slot

Let $\vec{a} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + \sqrt{2}\hat{k}$, $\vec{c} = 5\hat{i} + \hat{j} + \sqrt{2}\hat{k}$ be three vectors such that the projection vector of $\vec{b}$ on $\vec{a}$ is $\vec{a}$.

If $\vec{a} + \vec{b}$ is perpendicular to $\vec{c}$, then $|\vec{b}|$ is equal to :

(A) $\sqrt{32}$ (B) 6 (C) $\sqrt{22}$ (D) 4

Q140. JEE Main 2018 (Online) 16th April Morning Slot

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{c} = \hat{j} - \hat{k}$ and a vector $\vec{b}$ be such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 3$. Then $|\vec{b}|$ equals :

(A) $\frac{11}{3}$ (B) $\frac{11}{\sqrt{3}}$ (C) $\sqrt{\frac{11}{3}}$ (D) $\frac{\sqrt{11}}{3}$

Q141. JEE Main 2018 (Offline)

Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u} \cdot \vec{b} = 24$, then $|\vec{u}|^2$ is equal to

(A) 336 (B) 315 (C) 256 (D) 84

Q142. JEE Main 2018 (Online) 15th April Evening Slot

If the position vectors of the vertices A, B and C of a Δ ABC are respectively $4\hat{i} + 7\hat{j} + 8\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}$, and $2\hat{i} + 5\hat{j} + 7\hat{k}$, then the position vectors of the point, where the bisector of $\angle$A meets BC is :

(A) $\frac{1}{2}\left(4\hat{i} + 8\hat{j} + 11\hat{k}\right)$ (B) $\frac{1}{3}\left(6\hat{i} + 11\hat{j} + 15\hat{k}\right)$

(C) $\frac{1}{3}\left(6\hat{i} + 13\hat{j} + 18\hat{k}\right)$ (D) $\frac{1}{4}\left(8\hat{i} + 14\hat{j} + 19\hat{k}\right)$

Q143. JEE Main 2018 (Online) 15th April Morning Slot

If $\vec{a}$, $\vec{b}$, and $\vec{C}$ are unit vectors such that $\vec{a} + 2\vec{b} + 2\vec{c} = \vec{0}$, then $|\vec{a} \times \vec{c}|$ is equal to :

(A) $\frac{\sqrt{15}}{4}$ (B) $\frac{1}{4}$ (C) $\frac{15}{16}$ (D) $\frac{\sqrt{15}}{16}$

Q144. JEE Main 2017 (Online) 9th April Morning Slot

If the vector $\vec{b} = 3\hat{j} + 4\hat{k}$ is written as the sum of a vector $\vec{b_1}$, parallel to $\vec{a} = \hat{i} + \hat{j}$ and a vector $\vec{b_2}$, perpendicular to $\vec{a}$, then $\vec{b_1} \times \vec{b_2}$ is equal to :

(A) $-3\hat{i} + 3\hat{j} - 9\hat{k}$ (B) $6\hat{i} - 6\hat{j} + \frac{9}{2}\hat{k}$

(C) $-6\hat{i} + 6\hat{j} - \frac{9}{2}\hat{k}$ (D) $3\hat{i} - 3\hat{j} + 9\hat{k}$

Q145. JEE Main 2017 (Online) 8th April Morning Slot

The area (in sq. units) of the parallelogram whose diagonals are along the vectors $8\hat{i} - 6\hat{j}$ and $3\hat{i} + 4\hat{j} - 12\hat{k}$, is :

(A) 26 (B) 65 (C) 20 (D) 52

Q146. JEE Main 2017 (Offline)

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$.

Let $\vec{c}$ be a vector such that $|\vec{c} - \vec{a}| = 3$,

$\left|\left(\vec{a} \times \vec{b}\right) \times \vec{c}\right| = 3$ and the angle between $\vec{c}$ and $\vec{a} \times \vec{b}$ is 30°.

Then $\vec{a}.\vec{c}$ is equal to :

(A) 2 (B) 5 (C) $\frac{1}{8}$ (D) $\frac{25}{8}$

Q147. JEE Main 2016 (Online) 10th April Morning Slot

Let ABC be a triangle whose circumcentre is at P. If the position vectors of A, B, C and P are $\vec{a}, \vec{b}, \vec{c}$ and $\frac{\vec{a}+\vec{b}+\vec{c}}{4}$ respectively, then the position vector of the orthocentre of this triangle, is :

(A) $\vec{a} + \vec{b} + \vec{c}$

(B) $-\left(\frac{\vec{a}+\vec{b}+\vec{c}}{2}\right)$

(C) $\vec{0}$

(D) $\left(\frac{\vec{a}+\vec{b}+\vec{c}}{2}\right)$

Q148. JEE Main 2016 (Online) 9th April Morning Slot

In a triangle ABC, right angled at the vertex A, if the position vectors of A, B and C are respectively $3\hat{i}$ + $\hat{j} - \hat{k}$, $-\hat{i}$ + $3\hat{j}$ + p$\hat{k}$ and $5\hat{i}$ + q$\hat{j} - 4\hat{k}$, then the point (p, q) lies on a line :

(A) parallel to x-axis.

(B) parallel to y-axis.

(C) making an acute angle with the positive direction of x-axis.

(D) making an obtuse angle with the positive direction of x-axis.

Q149. JEE Main 2013 (Offline)

If the vectors $\overrightarrow{AB} = 3\hat{i} + 4\hat{k}$ and $\overrightarrow{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangle ABC, then the length of the median through A is :

(A) $\sqrt{18}$ (B) $\sqrt{72}$ (C) $\sqrt{33}$ (D) $\sqrt{45}$

Q150. AIEEE 2012

Let $\vec{a}$ and $\vec{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q151. AIEEE 2012

Let $ABCD$ be a parallelogram such that $\overrightarrow{AB} = \vec{q}, \overrightarrow{AD} = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincide with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by :

(A) $\vec{r} = 3\vec{q} - \frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(B) $\vec{r} = -\vec{q} + \frac{(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(C) $\vec{r} = \vec{q} - \frac{(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

(D) $\vec{r} = -3\vec{q} - \frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}$

Q152. AIEEE 2011

The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a}.\vec{d} = 0$. Then the vector $\vec{d}$ is equal to :

(A) $\vec{c} + \left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

(B) $\vec{b} + \left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$

(C) $\vec{c} - \left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

(D) $\vec{b} - \left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$

Q153. AIEEE 2011

Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a} + 3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a} + 3\vec{b} + 6\vec{c}$ is :

(A) $\vec{a} + \vec{c}$

(B) $\vec{c}$

(C) $\vec{a}$

(D) $\vec{0}$

Q154. AIEEE 2010

If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then (λ, μ) is equal to :

(A) $(2, -3)$ (B) $(-2, 3)$ (C) $(3, -2)$ (D) $(-3, 2)$

Q155. AIEEE 2008

The non-zero vectors are $\vec{a}, \vec{b}$ and $\vec{c}$ are related by $\vec{a} = 8\vec{b}$ and $\vec{c} = -7\vec{b}$. Then the angle between $\vec{a}$ and $\vec{c}$ is :

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

Q156. AIEEE 2007

If $\hat{u}$ and $\hat{v}$ are unit vectors and θ is the acute angle between them, then $2\hat{u} \times 3\hat{v}$ is a unit vector for :

(A) no value of θ (B) exactly one value of θ

(C) exactly two values of θ (D) more than two values of θ

Q157. AIEEE 2006

The values of a, for which the points A, B, C with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $a\hat{i} - 3\hat{j} + \hat{k}$ respectively are the vertices of a right angled triangle with $C = \frac{\pi}{2}$ are :

(A) 2 and 1 (B) -2 and -1 (C) -2 and 1 (D) 2 and -1

Q158. AIEEE 2005

If C is the mid point of AB and P is any point outside AB, then :

(A) $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$

(B) $\overrightarrow{PA} + \overrightarrow{PB} = \overrightarrow{PC}$

(C) $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC} = \vec{0}$

(D) $\overrightarrow{PA} + \overrightarrow{PB} = \overrightarrow{PC} = \vec{0}$

Q159. AIEEE 2005

Let a, b and c be distinct non-negative numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane, then c is :

(A) the Geometric Mean of a and b (B) the Arithmetic Mean of a and b

(C) equal to zero (D) the Harmonic Mean of a and b

Q160. AIEEE 2005

For any vector $\vec{a}$, the value of $\left(\vec{a}\times\hat{i}\right)^2+\left(\vec{a}\times\hat{j}\right)^2+\left(\vec{a}\times\hat{k}\right)^2$ is equal to :

(A) $3\vec{a}^2$ (B) $\vec{a}^2$ (C) $2\vec{a}^2$ (D) $4\vec{a}^2$

Q161. AIEEE 2004

A particle acted on by constant forces $4\hat{i}+\hat{j}-3\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ is displaced from the point $\hat{i}+2\hat{j}+3\hat{k}$ to the point $5\hat{i}+4\hat{j}+\hat{k}$. The total work done by the forces is :

(A) 50 units (B) 20 units (C) 30 units (D) 40 units

Q162. AIEEE 2004

Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of these are collinear. If the vector $\vec{a}+2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+3\vec{c}$ is collinear with $\vec{a}$ (λ being some non-zero scalar) then $\vec{a}+2\vec{b}+6\vec{c}$ equals to :

(A) $\vec{0}$ (B) $\lambda\vec{b}$ (C) $\lambda\vec{c}$ (D) $\lambda\vec{a}$

Q163. AIEEE 2004

Let $\vec{u},\vec{v},\vec{w}$ be such that $\left|\vec{u}\right|=1,\ \left|\vec{v}\right|=2,\ \left|\vec{w}\right|=3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and $\vec{v},\vec{w}$ are perpendicular to each other then $\left|\vec{u}-\vec{v}+\vec{w}\right|$ equals :

(A) 14 (B) $\sqrt{7}$ (C) $\sqrt{14}$ (D) 2

Q164. AIEEE 2004

Let $\vec{a},\vec{b}$ and $\vec{c}$ be non-zero vectors such that $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\frac{1}{3}\left|\vec{b}\right|\left|\vec{c}\right|\vec{a}$. If θ is the acute angle between the vectors $\vec{b}$ and $\vec{c}$, then $\sin\theta$ equals :

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Q165. AIEEE 2003

If $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$ then $\vec{a}+\vec{b}+\vec{c}=$

(A) abc (B) -1 (C) 0 (D) 2

Q166. AIEEE 2003

Let $\vec{u}=\hat{i}+\hat{j},\ \vec{v}=\hat{i}-\hat{j}$ and $\vec{w}=\hat{i}+2\hat{j}+3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u}.\hat{n}=0$ and $\vec{v}.\hat{n}=0$, then $\left|\vec{w}.\hat{n}\right|$ is equal to :

(A) 3 (B) 0 (C) 1 (D) 2

Q167. AIEEE 2003

The vectors $\overrightarrow{AB} = 3\hat{i} + 4\hat{k}$ & $\overrightarrow{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of triangle ABC. The length of the median through A is :

(A) $\sqrt{288}$ (B) $\sqrt{18}$ (C) $\sqrt{72}$ (D) $\sqrt{33}$

Q168. AIEEE 2003

$\vec{a}, \vec{b}, \vec{c}$ are 3 vectors, such that

$\vec{a} + \vec{b} + \vec{c} = 0$, $|\vec{a}| = 1$ $|\vec{b}| = 2$, $|\vec{c}| = 3$,,

then $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}$ is equal to :

(A) 1 (B) 0 (C) -7 (D) 7

Q169. AIEEE 2003

A tetrahedron has vertices at $O(0, 0, 0)$, $A(1, 2, 1)$, $B(2, 1, 3)$ and $C(-1, 1, 2)$. Then the angle between the faces OAB and ABC will be :

(A) 90° (B) $\cos^{-1}\left(\frac{19}{35}\right)$

(C) $\cos^{-1}\left(\frac{17}{31}\right)$ (D) 30°

Q170. AIEEE 2003

If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2)$,

$(1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then the product abc equals :

(A) 0 (B) 2 (C) -1 (D) 1

Q171. AIEEE 2003

Consider points A, B, C and D with position

vectors $7\hat{i} - 4\hat{j} + 7\hat{k}, \hat{i} - 6\hat{j} + 10\hat{k}, -\hat{i} - 3\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 5\hat{k}$ respectively. Then $ABCD$ is a :

(A) parallelogram but not a rhombus (B) square

(C) rhombus (D) None

Q172. AIEEE 2002

If $\left|\vec{a}\right| = 4, \left|\vec{b}\right| = 2$ and the angle between $\vec{a}$ and $\vec{b}$ is $\pi/6$ then $\left(\vec{a} \times \vec{b}\right)^2$ is equal to :

(A) 48 (B) 16 (C) $\vec{a}$ (D) none of these

Q173. AIEEE 2002

If the vectors $\vec{\mathbf{a}}, \vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ form the sides BC, CA and AB respectively of a triangle ABC, then :

(A) $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = \vec{\mathbf{b}} \cdot \vec{\mathbf{c}} = \vec{\mathbf{c}} \cdot \vec{\mathbf{b}} = 0$

(B) $\vec{\mathbf{a}} \times \vec{\mathbf{b}} = \vec{\mathbf{b}} \times \vec{\mathbf{c}} = \vec{\mathbf{c}} \times \vec{\mathbf{a}}$

(C) $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = \vec{\mathbf{b}} \cdot \vec{\mathbf{c}} = \vec{\mathbf{c}} \cdot \vec{\mathbf{a}} = 0$

(D) $\vec{\mathbf{a}} \times \vec{\mathbf{b}} + \vec{\mathbf{b}} \times \vec{\mathbf{c}} + \vec{\mathbf{c}} \times \vec{\mathbf{a}} = \vec{\mathbf{0}}$

Q174. AIEEE 2002

If $\left|\vec{a}\right| = 5, \left|\vec{b}\right| = 4, \left|\vec{c}\right| = 3$ thus what will be the value of $\left|\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}\right|$, given that $\vec{a} + \vec{b} + \vec{c} = 0$:

(A) 25 (B) 50 (C) -25 (D) -50

Q175. AIEEE 2002

If the vectors $\vec{c}, \vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\hat{b} = \hat{j}$ are such that $\vec{a}, \vec{c}$ and $\vec{b}$ form a right handed system then $\vec{c}$ is :

(A) $z\hat{i} - x\hat{k}$ (B) $\vec{0}$ (C) $y\hat{j}$ (D) $-z\hat{i} + x\hat{k}$

Q176. AIEEE 2002

$\vec{a} = 3\hat{i} - 5\hat{j}$ and $\vec{b} = 6\hat{i} + 3\hat{j}$ are two vectors and $\vec{c}$ is a vector such that $\vec{c} = \vec{a} \times \vec{b}$ then $\left|\vec{a}\right| : \left|\vec{b}\right| : \left|\vec{c}\right|$ =

(A) $\sqrt{34} : \sqrt{45} : \sqrt{39}$ (B) $\sqrt{34} : \sqrt{45} : 39$

(C) $34 : 39 : 45$ (D) $39 : 35 : 34$

Numerical 47 questions

Q177. JEE Main 2026 (Online) 6th April Morning Shift

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{c} \cdot (\vec{a} - 2\vec{b})$ is equal to ____ .

Q178. JEE Main 2026 (Online) 28th January Morning Shift

Let PQR be a triangle such that $\overrightarrow{PQ} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $\overrightarrow{PR} = a\hat{i} + b\hat{j} - 4\hat{k}, a, b \in Z$. Let S be the point on QR , which is equidistant from the lines PQ and PR . If $|\overrightarrow{PR}| = 9$ and $\overrightarrow{PS} = \hat{i} - 7\hat{j} + 2\hat{k}$, then the value of $3a - 4b$ is ____

Q179. JEE Main 2026 (Online) 22nd January Evening Shift

Let a vector $\vec{a} = \sqrt{2}\hat{i} - \hat{j} + \lambda\hat{k}, \lambda > 0$, make an obtuse angle with the vector $\vec{b} = -\lambda^2\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$ and an angle $\theta, \frac{\pi}{6} < \theta < \frac{\pi}{2}$, with the positive z-axis. If the set of all possible values of λ is $(\alpha, \beta) - \{\gamma\}$, then $\alpha + \beta + \gamma$ is equal to ____ .

Q180. JEE Main 2025 (Online) 4th April Evening Shift

Let the three sides of a triangle ABC be given by the vectors $2\hat{i} - \hat{j} + \hat{k}, \hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$. Let G be the centroid of the triangle ABC. Then $6\left(|\overrightarrow{AG}|^2 + |\overrightarrow{BG}|^2 + |\overrightarrow{CG}|^2\right)$ is equal to _____.

Q181. JEE Main 2025 (Online) 3rd April Evening Shift

Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}, \vec{b} = 3\hat{i} - 3\hat{j} + 3\hat{k}, \vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 4$. Then $|(\vec{a} \times \vec{d})|^2$ is equal to_____.

Q182. JEE Main 2025 (Online) 3rd April Morning Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}, \vec{c} = \lambda\hat{j} + \mu\hat{k}$ and $\hat{d}$ be a unit vector such that $\vec{a} \times \hat{d} = \vec{b} \times \hat{d}$ and $\vec{c} \cdot \hat{d} = 1$. If $\vec{c}$ is perpendicular to $\vec{a}$, then $|3\lambda\hat{d} + \mu\vec{c}|^2$ is equal to_____

Q183. JEE Main 2025 (Online) 28th January Morning Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{d} = \vec{a} \times \vec{b}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|, |\vec{c} - 2\vec{a}|^2 = 8$ and the angle between $\vec{d}$ and $\vec{c}$ is $\frac{\pi}{4}$, then $|10 - 3\vec{b} \cdot \vec{c}| + |\vec{d} \times \vec{c}|^2$ is equal to _____.

Q184. JEE Main 2025 (Online) 22nd January Morning Shift

Let $\vec{c}$ be the projection vector of $\vec{b} = \lambda\hat{i} + 4\hat{k}, \lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

Q185. JEE Main 2024 (Online) 8th April Morning Shift

Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}, \vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____.

Q186. JEE Main 2024 (Online) 6th April Morning Shift

Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}, \vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$. If $\vec{a}\cdot\vec{c}=13$, then $(24-\vec{b}\cdot\vec{c})$ is equal to _____.

Q187. JEE Main 2024 (Online) 5th April Morning Shift

Let $\overrightarrow{\mathrm{a}}=\hat{i}-3\hat{j}+7\hat{k}, \overrightarrow{\mathrm{b}}=2\hat{i}-\hat{j}+\hat{k}$ and $\overrightarrow{\mathrm{c}}$ be a vector such that $(\overrightarrow{\mathrm{a}}+2\overrightarrow{\mathrm{b}})\times\overrightarrow{\mathrm{c}}=3(\overrightarrow{\mathrm{c}}\times\overrightarrow{\mathrm{a}})$. If $\vec{a}\cdot\vec{c}=130$, then $\vec{b}\cdot\vec{c}$ is equal to _____.

Q188. JEE Main 2024 (Online) 4th April Morning Shift

Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{\mathrm{AB}}=\hat{i}+2\hat{j}-7\hat{k}, \overrightarrow{\mathrm{BC}}=\mathrm{a}\hat{i}+\mathrm{b}\hat{j}+\mathrm{c}\hat{k}$ and $\overrightarrow{\mathrm{AC}}=6\hat{i}+\mathrm{d}\hat{j}-2\hat{k},\ \mathrm{d}>0$. Then the square of the length of the largest side of the triangle ABC is _____.

Q189. JEE Main 2024 (Online) 1st February Evening Shift

Let $\overrightarrow{\mathrm{a}}=\hat{i}+\hat{j}+\hat{k}, \overrightarrow{\mathrm{b}}=-\hat{i}-8\hat{j}+2\hat{k}$ and $\overrightarrow{\mathrm{c}}=4\hat{i}+\mathrm{c}_2\hat{j}+\mathrm{c}_3\hat{k}$ be three vectors such that $\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{a}}=\overrightarrow{\mathrm{c}}\times\overrightarrow{\mathrm{a}}$. If the angle between the vector $\overrightarrow{\mathrm{c}}$ and the vector $3\hat{i}+4\hat{j}+\hat{k}$ is θ, then the greatest integer less than or equal to $\tan^2\theta$ is _____.

Q190. JEE Main 2024 (Online) 31st January Evening Shift

Let $\vec{a}=3\hat{i}+2\hat{j}+\hat{k}, \vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a}+\vec{b})\times\vec{c}=2(\vec{a}\times\vec{b})+24\hat{j}-6\hat{k}$ and $(\vec{a}-\vec{b}+\hat{i})\cdot\vec{c}=-3$. Then $|\vec{c}|^2$ is equal to _____.

Q191. JEE Main 2024 (Online) 31st January Morning Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}|=1, |\vec{b}|=4$, and $\vec{a}\cdot\vec{b}=2$. If $\vec{c}=(2\vec{a}\times\vec{b})-3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α, then $192\sin^2\alpha$ is equal to _____.

Q192. JEE Main 2024 (Online) 27th January Morning Shift

The least positive integral value of α, for which the angle between the vectors $\alpha\hat{i}-2\hat{j}+2\hat{k}$ and $\alpha\hat{i}+2\alpha\hat{j}-2\hat{k}$ is acute, is _____.

Q193. JEE Main 2023 (Online) 13th April Morning Shift

Let $\vec{a}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a}=\vec{b}\times\vec{c}$ and $|\vec{b}|^2=50$, then $|72-|\vec{b}+\vec{c}|^2|$ is equal to _____.

Q194. JEE Main 2023 (Online) 11th April Evening Shift

Let $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}-\hat{k}$. If $\vec{c}$ is a vector such that $\vec{a}\cdot\vec{c}=11, \vec{b}\cdot(\vec{a}\times\vec{c})=27$ and $\vec{b}\cdot\vec{c}=-\sqrt{3}|\vec{b}|$, then $|\vec{a}\times\vec{c}|^2$ is equal to _____.

Q195. JEE Main 2023 (Online) 8th April Morning Shift

Let $\vec{a}=6\hat{i}+9\hat{j}+12\hat{k}, \vec{b}=\alpha\hat{i}+11\hat{j}-2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a}\times\vec{c}=\vec{a}\times\vec{b}$. If

$\vec{a}\cdot\vec{c}=-12, \vec{c}\cdot(\hat{i}-2\hat{j}+\hat{k})=5$, then $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k})$ is equal to _____.

Q196. JEE Main 2023 (Online) 1st February Morning Shift

$A(2,6,2), B(-4,0,\lambda), C(2,3,-1)$ and $D(4,5,0), |\lambda|\leq 5$ are the vertices of a quadrilateral $ABCD$. If its area is 18 square units, then $5-6\lambda$ is equal to _____.

Q197. JEE Main 2023 (Online) 31st January Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that

$|\vec{a}|=\sqrt{31}, 4|\vec{b}|=|\vec{c}|=2$ and $2(\vec{a}\times\vec{b})=3(\vec{c}\times\vec{a})$.

If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left(\frac{\vec{a}\times\vec{c}}{\vec{a}\cdot\vec{b}}\right)^2$ is equal to _____.

Q198. JEE Main 2023 (Online) 31st January Morning Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}|=\sqrt{14}, |\vec{b}|=\sqrt{6}$ and $|\vec{a}\times\vec{b}|=\sqrt{48}$. Then $(\vec{a}\cdot\vec{b})^2$ is equal to _____.

Q199. JEE Main 2023 (Online) 24th January Evening Shift

Let $\overrightarrow{a}=\hat{i}+2\hat{j}+\lambda\hat{k}, \overrightarrow{b}=3\hat{i}-5\hat{j}-\lambda\hat{k}, \overrightarrow{a}.\overrightarrow{c}=7, 2\overrightarrow{b}.\overrightarrow{c}+43=0, \overrightarrow{a}\times\overrightarrow{c}=\overrightarrow{b}\times\overrightarrow{c}$. Then $\left|\overrightarrow{a}.\overrightarrow{b}\right|$ is equal to :

Q200. JEE Main 2022 (Online) 29th July Evening Shift

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}+\vec{b}|^2=|\vec{a}|^2+2|\vec{b}|^2, \vec{a}\cdot\vec{b}=3$ and $|\vec{a}\times\vec{b}|^2=75$. Then $|\vec{a}|^2$ is equal to _____.

Q201. JEE Main 2022 (Online) 27th July Evening Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{a} \times \vec{b} = 4\vec{c}, \vec{b} \times \vec{c} = 9\vec{a}$ and $\vec{c} \times \vec{a} = \alpha\vec{b}$, $\alpha > 0$. If $\left|\vec{a}\right| + \left|\vec{b}\right| + \left|\vec{c}\right| = \frac{1}{36}$, then α is equal to _____.

Q202. JEE Main 2022 (Online) 29th June Evening Shift

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} + \left(\vec{b} \times \vec{c}\right) = \vec{0}$ and $\vec{b} . \vec{c} = 5$. Then the value of $3\left(\vec{c} . \vec{a}\right)$ is equal to _____.

Q203. JEE Main 2022 (Online) 28th June Morning Shift

If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}, \vec{b} = 3\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ are coplanar vectors and $\vec{a} . \vec{c} = 5, \vec{b} \perp \vec{c}$, then $122(c_1 + c_2 + c_3)$ is equal to _____.

Q204. JEE Main 2022 (Online) 25th June Evening Shift

Let $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\lambda \in$ R. If $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} . \vec{b} + 21 = 0$, then $\left(\vec{b} - \vec{a}\right) . \left(\hat{k} - \hat{j}\right) + \left(\vec{b} + \vec{a}\right) . \left(\hat{i} - \hat{k}\right)$ is equal to _____.

Q205. JEE Main 2022 (Online) 25th June Morning Shift

Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}| = 4$, $|\vec{b}| = 3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then $\left|\left(\vec{a} - \vec{b}\right) \times \left(\vec{a} + \vec{b}\right)\right|^2 + 4\left(\vec{a} . \vec{b}\right)^2$ is equal to _____.

Q206. JEE Main 2021 (Online) 1st September Evening Shift

Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. Let a vector $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{v}$ is perpendicular to the vector $3\hat{i} + 2\hat{j} - \hat{k}$ and its projection on $\vec{a}$ is 19 units, then $\left|2\vec{v}\right|^2$ is equal to _____.

Q207. JEE Main 2021 (Online) 27th August Morning Shift

Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}, \vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $\left|\vec{b} \times \vec{c}\right| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $\left|\vec{a}\right|^2$ is _____.

Q208. JEE Main 2021 (Online) 26th August Evening Shift

If the projection of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then λ is equal to _____.

Q209. JEE Main 2021 (Online) 27th July Morning Shift

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b}$ and $\vec{c} = \hat{j} - \hat{k}$ be three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} . \vec{b} = 1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a} \times \vec{c}$ is l, then the value of $3l^2$ is equal to _____.

Q210. JEE Main 2021 (Online) 25th July Evening Shift

If $\left(\vec{a} + 3\vec{b}\right)$ is perpendicular to $\left(7\vec{a} - 5\vec{b}\right)$ and $\left(\vec{a} - 4\vec{b}\right)$ is perpendicular to $\left(7\vec{a} - 2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is _____.

Q211. JEE Main 2021 (Online) 25th July Morning Shift

Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$, and $\left|\vec{r}\right| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.

Q212. JEE Main 2021 (Online) 20th July Evening Shift

For p > 0, a vector $\vec{v_2} = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\vec{v_1} = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan\theta = \frac{\left(\alpha\sqrt{3}-2\right)}{\left(4\sqrt{3}+3\right)}$, then the value of α is equal to _____.

Q213. JEE Main 2021 (Online) 20th July Morning Shift

Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ, with the vector $\vec{a} + \vec{b} + \vec{c}$. Then $36\cos^2 2\theta$ is equal to _____.

Q214. JEE Main 2021 (Online) 17th March Evening Shift

Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If the vector $\vec{x}$ is perpendicular to $\left(3\hat{i} + 2\hat{j} - \hat{k}\right)$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to _____.

Q215. JEE Main 2021 (Online) 25th February Evening Shift

Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} . \vec{b}$ is equal to _____.

Q216. JEE Main 2021 (Online) 25th February Morning Shift

Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}, \vec{b} = \hat{i} - \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r}.\vec{b} = 0$, then $\vec{r}.\vec{a}$ is equal to _____.

Q217. JEE Main 2020 (Online) 6th September Evening Slot

If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $\left|\vec{x} + \vec{y}\right| = \left|\vec{x}\right|$ and $2\vec{x} + \lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is _____ .

Q218. JEE Main 2020 (Online) 6th September Morning Slot

If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of

$\sqrt{3}\left|\vec{a} + \vec{b}\right| + \left|\vec{a} - \vec{b}\right|$ is_____.

Q219. JEE Main 2020 (Online) 5th September Evening Slot

Let the vectors $\vec{a}, \vec{b}, \vec{c}$ be such that
$\left|\vec{a}\right| = 2, \left|\vec{b}\right| = 4$ and $\left|\vec{c}\right| = 4$. If the projection of
$\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$
and $\vec{b}$ is perpendicular to $\vec{c}$, then the value of
$\left|\vec{a} + \vec{b} - \vec{c}\right|$ is _____.

Q220. JEE Main 2020 (Online) 4th September Evening Slot

If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of

$\left|\hat{i} \times \left(\vec{a} \times \hat{i}\right)\right|^2 + \left|\hat{j} \times \left(\vec{a} \times \hat{j}\right)\right|^2 + \left|\hat{k} \times \left(\vec{a} \times \hat{k}\right)\right|^2$ is equal to____

Q221. JEE Main 2020 (Online) 2nd September Evening Slot

Let the position vectors of points 'A' and 'B' be
$\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + 3\hat{k}$, respectively. A point 'P' divides the line segment AB internally in the ratio λ : 1 ($\lambda > 0$). If O is the origin and
$\overrightarrow{OB}.\overrightarrow{OP} - 3\left|\overrightarrow{OA} \times \overrightarrow{OP}\right|^2 = 6$, then λ is equal to_____.

Q222. JEE Main 2020 (Online) 2nd September Morning Slot

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\left|\vec{a}-\vec{b}\right|^2 + \left|\vec{a}-\vec{c}\right|^2 = 8$.

Then $\left|\vec{a}+2\vec{b}\right|^2 + \left|\vec{a}+2\vec{c}\right|^2$ is equal to _____.

Q223. JEE Main 2020 (Online) 9th January Evening Slot

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\left|\vec{a}\right| = \sqrt{3}$, $\left|\vec{b}\right| = 5, \vec{b}.\vec{c} = 10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b}\times\vec{c}$, then $\left|\vec{a}\times\left(\vec{b}\times\vec{c}\right)\right|$ is equal to _____.

Out of syllabus 58 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q224. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i}-\hat{j}+\hat{k}, \hat{i}+2\hat{j}+\mu\hat{k}$ and $3\hat{i}-4\hat{j}+5\hat{k}$, where $\lambda-\mu=5$, are coplanar, then $\sum\limits_{(\lambda,\mu)\in S} 80\left(\lambda^2+\mu^2\right)$ is equal to :

(A) 2370 (B) 2130 (C) 2210 (D) 2290

Q225. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let a, b, c be three distinct real numbers, none equal to one. If the vectors $a\hat{i}+\hat{j}+\hat{k}, \hat{i}+b\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+c\hat{k}$ are coplanar, then $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ is equal to :

(A) -2 (B) 1 (C) -1 (D) 2

Q226. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let $\lambda\in\mathbb{Z}, \vec{a}=\lambda\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=3\hat{i}-\hat{j}+2\hat{k}$. Let $\vec{c}$ be a vector such that $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=\vec{0}, \vec{a}\cdot\vec{c}=-17$ and $\vec{b}\cdot\vec{c}=-20$. Then $|\vec{c}\times(\lambda\hat{i}+\hat{j}+\hat{k})|^2$ is equal to :

(A) 53 (B) 62 (C) 49 (D) 46

Q227. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

If four distinct points with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar, then $[\vec{a}\ \vec{b}\ \vec{c}]$ is equal to :

(A) $[\vec{d}\ \ \vec{b}\ \ \vec{a}]+[\vec{a}\ \ \vec{c}\ \ \vec{d}]+[\vec{d}\ \ \vec{b}\ \ \vec{c}]$

(B) $[\vec{b}\ \ \vec{c}\ \ \vec{d}]+[\vec{d}\ \ \vec{a}\ \ \vec{c}]+[\vec{d}\ \ \vec{b}\ \ \vec{a}]$

(C) $[\vec{a}\ \ \vec{d}\ \ \vec{b}]+[\vec{d}\ \ \vec{c}\ \ \vec{a}]+[\vec{d}\ \ \vec{b}\ \ \vec{c}]$

(D) $[\vec{d}\ \ \vec{c}\ \ \vec{a}]+[\vec{b}\ \ \vec{d}\ \ \vec{a}]+[\vec{c}\ \ \vec{d}\ \ \vec{b}]$

Q228. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let the vectors $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}, \vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$ and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}, \vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$ are also coplanar, then $6(\mathrm{a}+\mathrm{b}+\mathrm{c})$ is equal to :

(A) 12 (B) 6 (C) 0 (D) 4

Q229. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

Let the vectors $\vec{a}, \vec{b}, \vec{c}$ represent three coterminous edges of a parallelopiped of volume V. Then the volume of the parallelopiped, whose coterminous edges are represented by $\vec{a}, \vec{b} + \vec{c}$ and $\vec{a} + 2\vec{b} + 3\vec{c}$ is equal to :

(A) 3 V (B) 2 V (C) 6 V (D) V

Q230. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

The sum of all values of α, for which the points whose position vectors are $\hat{i} - 2\hat{j} + 3\hat{k}, 2\hat{i} - 3\hat{j} + 4\hat{k}, (\alpha+1)\hat{i} + 2\hat{k}$ and $9\hat{i} + (\alpha-8)\hat{j} + 6\hat{k}$ are coplanar, is equal to :

(A) 6 (B) 4 (C) -2 (D) 2

Q231. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

Let the position vectors of the points A, B, C and D be $5\hat{i} + 5\hat{j} + 2\lambda\hat{k}, \hat{i} + 2\hat{j} + 3\hat{k}, -2\hat{i} + \lambda\hat{j} + 4\hat{k}$ and $-\hat{i} + 5\hat{j} + 6\hat{k}$. Let the set $S = \{\lambda \in \mathbb{R} :$ the points A, B, C and D are coplanar $\}$.

Then $\sum\limits_{\lambda \in S} (\lambda + 2)^2$ is equal to :

(A) $\frac{37}{2}$ (B) 25 (C) 13 (D) 41

Q232. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Let $\lambda \in \mathbb{R}, \vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}, \vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$.

If $((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$,

then $|\lambda(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|^2$ is equal to :

(A) 136 (B) 140 (C) 144 (D) 132

Q233. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a} = \alpha\vec{b} - \hat{n}, (\alpha \neq 0)$ and $\vec{b} \cdot \vec{c} = 12$, then $\left|\vec{c} \times (\vec{a} \times \vec{b})\right|$ is equal to :

(A) 15 (B) 9 (C) 6 (D) 12

Q234. JEE Main 2023 (Online) 25th January Evening Shift · MCQ (Single Correct Answer)

Let $\vec{a} = -\hat{i} - \hat{j} + \hat{k}, \vec{a} . \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{i} - \hat{j}$. Then $\vec{a} - 6\vec{b}$ is equal to :

(A) $3\left(\hat{i} + \hat{j} + \hat{k}\right)$ (B) $3\left(\hat{i} - \hat{j} - \hat{k}\right)$

(C) $3\left(\hat{i} + \hat{j} - \hat{k}\right)$ (D) $3\left(\hat{i} - \hat{j} + \hat{k}\right)$

Q235. JEE Main 2023 (Online) 25th January Evening Shift · MCQ (Single Correct Answer)

If the four points, whose position vectors are $3\hat{i} - 4\hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} - \hat{k}, -2\hat{i} - \hat{j} + 3\hat{k}$ and $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$ are coplanar, then α is equal to :

(A) $\frac{73}{17}$ (B) $-\frac{73}{17}$ (C) $-\frac{107}{17}$ (D) $\frac{107}{17}$

Q236. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b} . \vec{c}$ = 0 and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2}$. If $\vec{d}$ be a vector such that $\vec{b} . \vec{d} = \vec{a} . \vec{b}$, then $(\vec{a} \times \vec{b}) . (\vec{c} \times \vec{d})$ is equal to

(A) $\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

Q237. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

Let $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}, \vec{v} = 2\hat{i} + \hat{j} - \hat{k}, \vec{v} . \vec{w} = 2$ and $\vec{v} \times \vec{w} = \vec{u} + \lambda\vec{v}$. Then $\vec{u} . \vec{w}$ is equal to :

(A) $-\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Q238. JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is :

(A) -5 (B) 5 (C) 1 (D) -1

Q239. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a} = 2\hat{i} - \hat{j} + 5\hat{k}$ and $\vec{b} = \alpha\hat{i} + \beta\hat{j} + 2\hat{k}$. If $((\vec{a} \times \vec{b}) \times \hat{i}) \cdot \hat{k} = \frac{23}{2}$, then $|\vec{b} \times 2\hat{j}|$ is equal to :

(A) 4 (B) 5 (C) $\sqrt{21}$ (D) $\sqrt{17}$

Q240. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is :

(A) $\frac{3\pi}{4}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{5}$ (D) $\frac{5\pi}{6}$

Q241. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

Let a vector $\vec{c}$ be coplanar with the vectors $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. If the vector $\vec{c}$ also satisfies the conditions $\vec{c}.\left[\left(\vec{a}+\vec{b}\right) \times \left(\vec{a}\times\vec{b}\right)\right] = -42$ and $\left(\vec{c}\times\left(\vec{a}-\vec{b}\right)\right).\hat{k} = 3$, then the value of $|\vec{c}|^2$ is equal to :

(A) 24 (B) 29 (C) 35 (D) 42

Q242. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

If $\vec{a}.\vec{b} = 1$, $\vec{b}.\vec{c} = 2$ and $\vec{c}.\vec{a} = 3$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right), \vec{b}\times\left(\vec{c}\times\vec{a}\right), \vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ is :

(A) 0

(B) $-6\vec{a}.\left(\vec{b}\times\vec{c}\right)$

(C) $-12\vec{c}.\left(\vec{a}\times\vec{b}\right)$

(D) $-12\vec{b}.\left(\vec{c}\times\vec{a}\right)$

Q243. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies

$\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\} + \vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\} + \vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\} = \vec{0}$, then $\vec{r}$ is equal to :

(A) $\frac{1}{3}(\vec{a}+\vec{b}+\vec{c})$ (B) $\frac{1}{3}(2\vec{a}+\vec{b}-\vec{c})$

(C) $\frac{1}{2}(\vec{a}+\vec{b}+\vec{c})$ (D) $\frac{1}{2}(\vec{a}+\vec{b}+2\vec{c})$

Q244. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a}\times\vec{c} = \vec{b}$ and $\vec{a}.\vec{c} = 3$, then $\vec{a}.(\vec{b}\times\vec{c})$ is equal to :

(A) -2 (B) -6 (C) 6 (D) 2

Q245. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b}\times(\vec{b}\times\vec{c})$. If magnitudes of the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are $\sqrt{2}$, 1 and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is $\theta\left(0 < \theta < \frac{\pi}{2}\right)$, then the value of 1 + tanθ is equal to :

(A) $\sqrt{3}+1$ (B) 2

(C) 1 (D) $\frac{\sqrt{3}+1}{\sqrt{3}}$

Q246. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Let the vectors

$(2+a+b)\hat{i}+(a+2b+c)\hat{j}-(b+c)\hat{k}, (1+b)\hat{i}+2b\hat{j}-b\hat{k}$ and $(2+b)\hat{i}+2b\hat{j}+(1-b)\hat{k}, a,b,c \in R$

be co-planar. Then which of the following is true?

(A) 2b = a + c (B) 3c = a + b (C) a = b + 2c (D) 2a = b + c

Q247. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Let a vector $\vec{a}$ be coplanar with vectors $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d}=3\hat{i}+2\hat{j}+6\hat{k}$, and $|\vec{a}|=\sqrt{10}$. Then a possible value of $[\vec{a}\ \vec{b}\ \vec{c}]+[\vec{a}\ \vec{b}\ \vec{d}]+[\vec{a}\ \vec{c}\ \vec{d}]$ is equal to :

(A) -42 (B) -40 (C) -29 (D) -38

Q248. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Let three vectors $\vec{a},\vec{b}$ and $\vec{c}$ be such that $\vec{a}\times\vec{b}=\vec{c},\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{a}|=2$. Then which one of the following is not true?

(A) $\vec{a}\times\left((\vec{b}+\vec{c})\times(\vec{b}-\vec{c})\right)=\vec{0}$

(B) Projection of $\vec{a}$ on $(\vec{b}\times\vec{c})$ is 2

(C) $[\vec{a}\ \vec{b}\ \vec{c}]+[\vec{c}\ \vec{a}\ \vec{b}]=8$

(D) $|3\vec{a}+\vec{b}-2\vec{c}|^2=51$

Q249. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}.\vec{c}=|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $(\vec{a}\times\vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is :

(A) $\frac{2}{3}$ (B) 4 (C) 3 (D) $\frac{3}{2}$

Q250. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

Let $\vec{a}$ = $2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}$ = $7\hat{i}+\hat{j}-6\hat{k}$.

If $\vec{r}\times\vec{a}=\vec{r}\times\vec{b}, \vec{r}.(\hat{i}+2\hat{j}+\hat{k})=-3$, then $\vec{r}.(2\hat{i}-3\hat{j}+\hat{k})$ is equal to :

(A) 10 (B) 8 (C) 13 (D) 12

Q251. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

If $\vec{a}$ and $\vec{b}$ are perpendicular, then
$\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \vec{b}\right)\right)\right)$ is equal to :

(A) $\frac{1}{2}|\vec{a}|^4\vec{b}$

(B) $\vec{0}$

(C) $\vec{a} \times \vec{b}$

(D) $|\vec{a}|^4\vec{b}$

Q252. JEE Main 2020 (Online) 5th September Morning Slot · MCQ (Single Correct Answer)

If the volume of a parallelopiped, whose coterminus edges are given by the vectors $\vec{a} = \hat{i} + \hat{j} + n\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and $\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}$ $(n \geq 0)$, is 158 cu. units, then :

(A) n = 7

(B) $\vec{b}.\vec{c} = 10$

(C) $\vec{a}.\vec{c} = 17$

(D) n = 9

Q253. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

Let x_0 be the point of Local maxima of $f(x) = \vec{a}.\left(\vec{b} \times \vec{c}\right)$, where $\vec{a} = x\hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{i} + x\hat{j} - \hat{k}, \vec{c} = 7\hat{i} - 2\hat{j} + x\hat{k}$. Then the value of $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}$ at $x = x_0$ is :

(A) 14

(B) -30

(C) -4

(D) -22

Q254. JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

Let the volume of a parallelopiped whose coterminous edges are given by

$\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}, \vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and

$\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos\theta$ can be :

(A) $\frac{7}{6\sqrt{3}}$

(B) $\frac{7}{6\sqrt{6}}$

(C) $\frac{5}{7}$

(D) $\frac{5}{3\sqrt{3}}$

Q255. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

Let $\alpha \in$ R and the three vectors

$\vec{a} = \alpha\hat{i} + \hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$

and $\vec{c} = \alpha\hat{i} - 2\hat{j} + 3\hat{k}$.

Then the set S = {$\alpha : \vec{a}, \vec{b}$ and $\vec{c}$ are coplanar} :

(A) contains exactly two numbers only one of which is positive

(B) is singleton

(C) contains exactly two positive numbers

(D) is empty

Q256. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the volume of parallelopiped formed by the vectors $\hat{i} + \lambda\hat{j} + \hat{k}$, $\hat{j} + \lambda\hat{k}$ and $\lambda\hat{i} + \hat{k}$ is minimum, then λ is equal to :

(A) $-\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $-\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

Q257. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The magnitude of the projection of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ on the vector perpendicular to the plane containing the vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$, is :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{6}$ (C) $\sqrt{\frac{3}{2}}$ (D) $3\sqrt{6}$

Q258. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors, out of which vectors $\vec{b}$ and $\vec{c}$ are non-parallel. If α and β are the angles which vector $\vec{a}$ makes with vectors $\vec{b}$ and $\vec{c}$ respectively and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2}\vec{b}$, then $|\alpha - \beta|$ is equal to :

(A) 90° (B) 30° (C) 45° (D) 60°

Q259. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

The sum of the distinct real values of μ, for which the vectors, $\mu\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + \mu\hat{j} + \hat{k}$, $\hat{i} + \hat{j} + \mu\hat{k}$ are co-planar, is :

(A) 2 (B) -1 (C) 0 (D) 1

Q260. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} - \hat{j}, \vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$

be a vector such that $\vec{a} \times \vec{c} + \vec{b} = \vec{0}$

and $\vec{a} \cdot \vec{c} = 4$, then $|\vec{c}|^2$ is equal to :

(A) 8 (B) $\frac{19}{2}$ (C) 9 (D) $\frac{17}{2}$

Q261. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} \times \left(\vec{b} \times \vec{c}\right) = \frac{\sqrt{3}}{2}\left(\vec{b} + \vec{c}\right)$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is:

(A) $\frac{2\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Q262. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and

$\left(\vec{a} \times \vec{b}\right) \times \vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$. If θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin\theta$ is :

(A) $\frac{2}{3}$ (B) $\frac{-2\sqrt{3}}{3}$ (C) $\frac{2\sqrt{2}}{3}$ (D) $\frac{-\sqrt{2}}{3}$

Q263. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

If $\left[\vec{a} \times \vec{b} \;\; \vec{b} \times \vec{c} \;\; \vec{c} \times \vec{a}\right] = \lambda\left[\vec{a} \;\; \vec{b} \;\; \vec{c}\right]^2$ then λ is equal to :

(A) 0 (B) 1 (C) 2 (D) 3

Q264. AIEEE 2011 · MCQ (Single Correct Answer)

If $\vec{a} = \frac{1}{\sqrt{10}}\left(3\hat{i} + \hat{k}\right)$ and $\vec{b} = \frac{1}{7}\left(2\hat{i} + 3\hat{j} - 6\hat{k}\right)$, then the value

of $\left(2\vec{a} - \vec{b}\right) \cdot \left[\left(\vec{a} \times \vec{b}\right) \times \left(\vec{a} + 2\vec{b}\right)\right]$ is :

(A) -3 (B) 5 (C) 3 (D) -5

Q265. AIEEE 2010 · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} + \vec{c} = \vec{0}$ and $\vec{a} \cdot \vec{b} = 3$:

(A) $2\hat{i} - \hat{j} + 2\hat{k}$ (B) $\hat{i} - \hat{j} - 2\hat{k}$

(C) $\hat{i} + \hat{j} - 2\hat{k}$ (D) $-\hat{i} + \hat{j} - 2\hat{k}$

Q266. AIEEE 2009 · MCQ (Single Correct Answer)

If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality $\left[3\vec{u}\ p\vec{v}\ p\vec{w}\right] - \left[p\vec{v}\ \vec{w}\ q\vec{u}\right] - \left[2\vec{w}\ q\vec{v}\ q\vec{u}\right] = 0$ holds for :

(A) exactly two values of (p, q)

(B) more than two but not all values of (p, q)

(C) all values of (p, q)

(D) exactly one value of (p, q)

Q267. AIEEE 2008 · MCQ (Single Correct Answer)

The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β ?

(A) $\alpha = 2,\ \beta = 2$

(B) $\alpha = 1,\ \beta = 2$

(C) $\alpha = 2,\ \beta = 1$

(D) $\alpha = 1,\ \beta = 1$

Q268. AIEEE 2007 · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = x\hat{i} + (x-2)\hat{j} - \hat{k}$. If the vectors $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{b}$, then x equals :

(A) -4

(B) -2

(C) 0

(D) 1.

Q269. AIEEE 2006 · MCQ (Single Correct Answer)

If $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = \vec{a} \times \left(\vec{b} \times \vec{c}\right)$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}.\vec{b} \neq 0,\ \vec{b}.\vec{c} \neq 0$ then $\vec{a}$ and $\vec{c}$ are :

(A) inclined at an angle of $\frac{\pi}{3}$ between them

(B) inclined at an angle of $\frac{\pi}{6}$ between them

(C) perpendicular

(D) parallel

Q270. AIEEE 2005 · MCQ (Single Correct Answer)

If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and λ is a real number then

$\left[\lambda\left(\vec{a}+\vec{b}\right) \quad \lambda^2\vec{b} \quad \lambda\vec{c}\right] = \left[\vec{a} \quad \vec{b}+\vec{c} \quad \vec{b}\right]$ for :

(A) exactly one value of λ

(B) no value of λ

(C) exactly three values of λ

(D) exactly two values of λ

Q271. AIEEE 2005 · MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$. Then $\left[\vec{a}, \vec{b}, \vec{c}\right]$ depends on :

(A) only y (B) only x (C) both x and y (D) neither x nor y

Q272. AIEEE 2004 · MCQ (Single Correct Answer)

If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and λ is a real number, then the vectors $\vec{a} + 2\vec{b} + 3\vec{c}$, $\lambda\vec{b} + 4\vec{c}$ and $(2\lambda - 1)\vec{c}$ are non coplanar for :

(A) no value of λ (B) all except one value of λ

(C) all except two values of λ (D) all values of λ

Q273. AIEEE 2003 · MCQ (Single Correct Answer)

If $\vec{u}, \vec{v}$ and $\vec{w}$ are three non-coplanar vectors, then $\left(\vec{u} + \vec{v} - \vec{w}\right).\left(\vec{u} - \vec{v}\right) \times \left(\vec{v} - \vec{w}\right)$ equals :

(A) $3\vec{u}.\vec{v} \times \vec{w}$ (B) 0

(C) $\vec{u}.\vec{v} \times \vec{w}$ (D) $\vec{u}.\vec{w} \times \vec{v}$

Q274. AIEEE 2002 · MCQ (Single Correct Answer)

If $\vec{a}$, $\vec{b}$, $\vec{c}$ are vectors such that $\left[\vec{a}\,\vec{b}\,\vec{c}\right] = 4$ then $\left[\vec{a} \times \vec{b}\;\vec{b} \times \vec{c}\;\vec{c} \times \vec{a}\right] =$

(A) 16 (B) 64 (C) 4 (D) 8

Q275. JEE Main 2023 (Online) 1st February Morning Shift · Numerical

Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}, \vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $\left[\vec{u}\;\vec{v}\;\vec{w}\right]$ is $-\alpha\sqrt{3401}$, and $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then $m + n$ is equal to _____.

Q276. JEE Main 2023 (Online) 29th January Morning Shift · Numerical

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a} - \vec{b} + \vec{c}, \lambda\vec{a} - 3\vec{b} + 4\vec{c}, -\vec{a} + 2\vec{b} - 3\vec{c}$ and $2\vec{a} - 4\vec{b} + 6\vec{c}$ respectively. If $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar, then λ is equal to _____.

Q277. JEE Main 2021 (Online) 27th July Evening Shift · Numerical

Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a}.\vec{b} = -1$ and $\vec{b}.\vec{c} = 10$, then $\left(\vec{a} \times \vec{b}\right).\vec{c}$ is equal to _____.

Q278. JEE Main 2021 (Online) 17th March Morning Shift · **Numerical**

If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$,

$\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and

$\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$

such that $\vec{a}.\vec{b} = 1$ and $\vec{b}.\vec{c} = -3$, then $\frac{1}{3}\left(\left(\vec{a}\times\vec{b}\right).\vec{c}\right)$ is equal to _____.

Q279. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

Let $\vec{c}$ be a vector perpendicular to the vectors, $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c}.\left(\hat{i} + \hat{j} + 3\hat{k}\right) = 8$ then the value of $\vec{c}.\left(\vec{a}\times\vec{b}\right)$ is equal to _____.

Q280. JEE Main 2021 (Online) 24th February Morning Shift · **Numerical**

Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a}.\vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2\left|\vec{a} + \vec{b} + \vec{c}\right|^2$ is _____.

Q281. JEE Main 2020 (Online) 9th January Morning Slot · **Numerical**

If the vectors, $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$,

$\vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and

$\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k}\,(a \in R)$

are coplanar and $3\left(\vec{p}.\vec{q}\right)^2 - \lambda\left|\vec{r}\times\vec{q}\right|^2 = 0$, then the value of λ is _____.

Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **B**	4. **A**	5. **A**
6. **B**	7. **B**	8. **C**	9. **B**	10. **B**
11. *Scan QR*	12. **C**	13. **A**	14. **C**	15. **D**
16. **B**	17. **D**	18. **A**	19. **A**	20. **C**
21. **D**	22. **A**	23. **A**	24. **B**	25. **D**
26. **A**	27. **D**	28. **C**	29. **C**	30. **C**
31. **D**	32. **B**	33. **D**	34. **A**	35. **C**
36. **A**	37. **D**	38. **D**	39. **D**	40. **C**
41. **A**	42. **C**	43. **B**	44. **B**	45. **B**
46. **C**	47. **C**	48. **A**	49. **C**	50. **A**
51. **D**	52. **A**	53. **C**	54. **C**	55. **A**
56. **A**	57. **B**	58. **B**	59. **C**	60. **D**
61. **D**	62. **B**	63. **C**	64. **D**	65. **C**
66. **D**	67. **A**	68. **C**	69. **C**	70. **D**
71. **A**	72. **B**	73. **D**	74. **D**	75. **C**
76. **B**	77. **B**	78. **D**	79. **B**	80. **C**
81. **B**	82. **A**	83. **D**	84. **C**	85. **C**
86. **A**	87. **A**	88. **C**	89. **D**	90. **A**
91. **D**	92. **B**	93. **D**	94. **C**	95. **A**
96. **B**	97. **C**	98. **B**	99. **D**	100. **D**
101. **A**	102. **C**	103. **A**	104. **A**	105. **D**
106. **C**	107. **D**	108. **A**	109. **D**	110. **B**
111. **C**	112. **C**	113. **C**	114. **D**	115. **B**
116. **D**	117. **A**	118. **C**	119. **D**	120. **C**
121. **D**	122. **B**	123. **D**	124. **B**	125. **A**
126. **D**	127. **A**	128. **C**	129. **B**	130. **A**
131. **A**	132. **C**	133. **B**	134. **C**	135. **B**
136. **B**	137. **D**	138. **C**	139. **B**	140. **C**
141. **A**	142. **C**	143. **A**	144. **B**	145. **B**
146. **A**	147. **D**	148. **C**	149. **C**	150. **C**

151. **B**	152. **C**	153. **D**	154. **D**	155. **D**
156. **B**	157. **A**	158. **A**	159. **A**	160. **C**
161. **D**	162. **A**	163. **C**	164. **A**	165. **C**
166. **A**	167. **D**	168. **C**	169. **B**	170. **C**
171. **D**	172. **B**	173. **B**	174. **A**	175. **A**
176. **B**				

Numerical

177. **3**	178. **37**	179. **5**	180. **164**	181. **128**
182. **5**	183. **6**	184. **16**	185. **569**	186. **46**
187. **30**	188. **54**	189. **38**	190. **38**	191. **48**
192. **5**	193. **66**	194. **285**	195. **11**	196. **11**
197. **3**	198. **36**	199. **8**	200. **14**	201. **36**
202. **BONUS**	203. **150**	204. **14**	205. **576**	206. **1494**
207. **90**	208. **5**	209. **2**	210. **60**	211. **3**
212. **6**	213. **4**	214. **486**	215. **2**	216. **12**
217. **1**	218. **4**	219. **6**	220. **18**	221. **0.8**
222. **2**	223. **30**			

Out of syllabus

224. **D**	225. **B**	226. **D**	227. **D**	228. **A**
229. **D**	230. **D**	231. **D**	232. **B**	233. **D**
234. **A**	235. **A**	236. **C**	237. **D**	238. **A**
239. **B**	240. **A**	241. **C**	242. **A**	243. **C**
244. **A**	245. **B**	246. **A**	247. **A**	248. **D**
249. **D**	250. **D**	251. **D**	252. **B**	253. **D**
254. **A**	255. **D**	256. **D**	257. **C**	258. **B**
259. **B**	260. **B**	261. **B**	262. **C**	263. **B**
264. **D**	265. **D**	266. **D**	267. **D**	268. **B**
269. **D**	270. **B**	271. **D**	272. **C**	273. **C**
274. **A**	275. **3501**	276. **2**	277. **9**	278. **2**
279. **28**	280. **75**	281. **1**		